

KNOWLEDGE INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

Approved by AICTE, Affiliated to Anna University, Chennai.

Accredited by NBA (CSE, ECE, EEE & MECH), Accredited by NAAC with "A" Grade KIOT Campus,
Kakapalayam (PO), Salem – 637 504, Tamil Nadu, India.



M.E. / M.Tech. / M.C.A. Regulations 2023

M.E. – Software Engineering

Semester – II to IV

Curriculum and Syllabi

(For the students admitted from the Academic Year 2025-26 Onwards)

Version: 1.0	Date: 15.12.2025
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KNOWLEDGE INSTITUTE OF TECHNOLOGY(AUTONOMOUS), SALEM -637504

Approved by AICTE, Affiliated to Anna University,
Accredited by NAAC and NBA (B.E.:Mech., ECE, EEE & CSE)

Website: www.kiot.ac.in

M.E. / M.Tech. / M.C.A. REGULATIONS 2023 (R 2023)
CHOICE BASED CREDIT SYSTEM AND OUTCOME BASED EDUCATION

M.E. SOFTWARE ENGINEERING

VISION OF THE INSTITUTE

To be a world class institution to impart value and need based professional education to the aspiring youth and carving them into disciplined world class professional who have the quest for excellence, achievement orientation and social responsibilities.

MISSION OF THE INSTITUTE

A	To promote academic growth by offering state-of-art undergraduate, postgraduate and doctoral programs and to generate new knowledge by engaging in cutting – edge research
B	To nurture talent, innovation, entrepreneurship, all-round personality and value system among the students and to foster competitiveness among students
C	To undertake collaborative projects which offer opportunities for long-term interaction with academia and industry
D	To pursue global standards of excellence in all our endeavors namely teaching, research, consultancy, continuing education and support functions

VISION OF THE DEPARTMENT

To develop Software Engineers and skilled professionals who can meet evolving industry demands and global challenges along with strong social values.

MISSION OF THE DEPARTMENT

M1	To empower aspiring software engineers with a comprehensive education, fostering innovation, critical thinking, and ethical leadership to drive technological advancements.
M2	To cultivate a diverse community of software engineers equipped with technical expertise, problem-solving abilities, and a commitment to creating impactful solutions that positively transform industries and communities.
M3	To prepare skilled graduates embracing lifelong learning and adaptability that are aligned with Sustainable Development Goals.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1	Identify the requirements and implement reliable, innovative and appropriate software solutions for the industrial need
PEO 2	Analyze and evaluate problems critically using the theoretical and technical knowledge to develop sustainable solutions and systems
PEO 3	Enhance skills through lifelong learning as software professionals to progress in managerial and leadership roles.

PROGRAM OUTCOMES (POs)	
Engineering Graduates will be able to:	
PO1	An ability to independently carry out research / investigation and development work to solve practical problems.
PO2	An ability to write and present a substantial technical report/document.
PO3	Students should be able to demonstrate a degree of mastery over the area of Software Engineering.
PO4	Collect requirements from the stakeholders and design software engineering applications with deep understanding of best software principles and practices.
PO5	Apply software testing techniques to produce error free and reliable software and ensure Quality.
PO6	Manage software project with state of the art approaches to ensure balance in all project areas like time, cost, quality, risk and human resource.



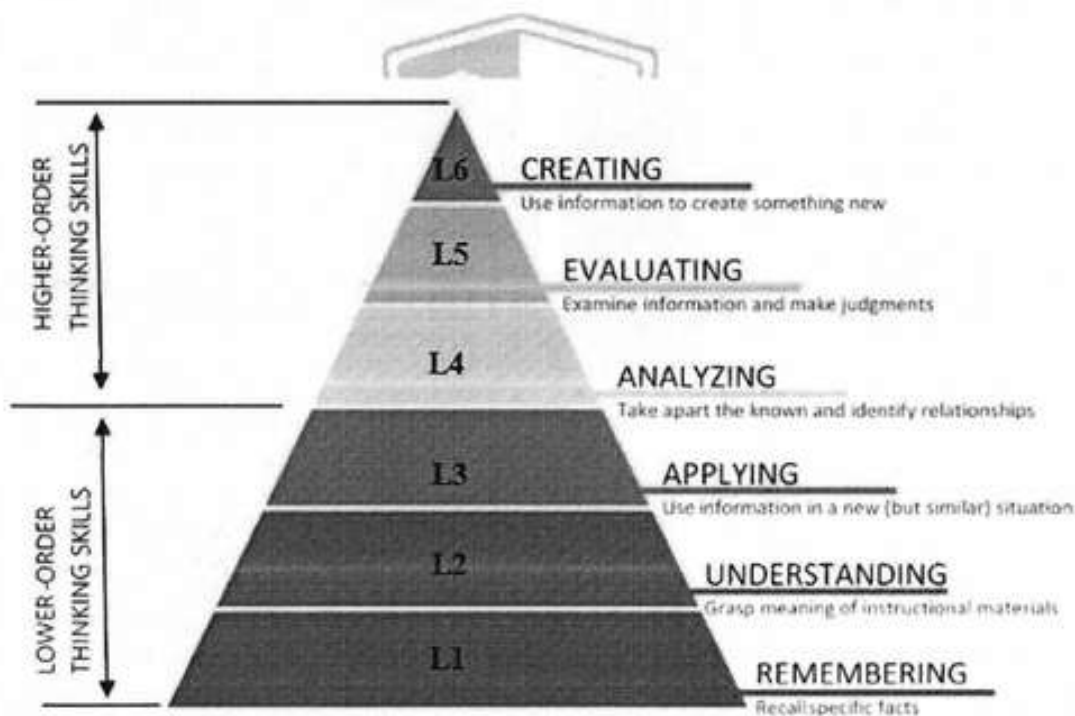

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BLOOM'S TAXONOMY LEVELS AND ACTION VERBS

(A) BLOOM'S TAXONOMY LEVELS (BTL):

Bloom's Taxonomy (BT) is based on the belief that learners must first acquire basic foundational knowledge about a subject before progressing to more complex types of thinking, such as analysis and evaluation. Bloom's Taxonomy levels help faculty to guide students through the learning process, from fundamental remembering and understanding to more complex evaluating and creating.



At KIOT, Curriculum Design, Delivery and Assessment (CDDA) are carried out based on the Blooms' Taxonomy Levels (BTL). Its organized set of objectives helps teachers to plan and deliver appropriate instruction, design valid assessment tasks and schemes, and ensure that instruction and assessment are aligned with the objectives.

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(B) BLOOM'S TAXONOMY ACTION VERBS

I. Remembering	II. Understanding	III. Applying	IV. Analyzing	V. Evaluating	VI. Creating
Exhibit memory of previously learned material by recalling facts, terms, basic concepts, and answers.	Demonstrate understanding of facts and ideas by organizing, comparing, interpreting, giving descriptions, and stating main ideas.	Solve problems to new situations by applying acquired knowledge, facts, techniques and rules in a different way.	Examine and break information into parts by identifying motives or causes. Make inferences and find evidence to support generalizations.	Present and defend opinions by making judgments about information, validity of ideas, or quality of work based on a set of criteria.	Compile information together in a different way by combining elements in a new pattern or proposing new Solutions.
- Choose - Define - Find - How - Label - List - Match - Name - Omit - Recall - Relate - Select - Show - Spell - Tell - What - When - Where - Which - Who - Why	- Classify - Compare - Contrast - Demonstrate - Explain - Extend - Illustrate - Infer - Interpret - Outline - Relate - Rephrase - Show - Summarize - Translate	- Apply - Build - Choose - Construct - Develop - Experiment with - Identify - Interview - Make use of - Model - Organize - Plan - Select - Solve - Utilize	- Analyze - Assume - Categorize - Classify - Compare - Conclusion - Contrast - Discover - Dissect - Distinguish - Divide - Examine - Function - Inference - Inspect - List - Motive - Relationships - Simplify - Survey - Take part in - Test for - Theme	- Agree - Appraise - Assess - Award - Choose - Compare - Conclude - Criteria - Criticize - Decide - Deduct - Defend - Determine - Disprove - Estimate - Evaluate - Explain - Importance - Influence - Interpret - Judge - Justify - Mark - Measure - Opinion - Perceive - Prioritize - Prove - Rate - Recommend - Rule on - Select - Support - Value	- Adapt - Build - Change - Choose - Combine - Compile - Compose - Construct - Create - Delete - Design - Develop - Discuss - Elaborate - Estimate - Formulate - Happen - Imagine - Improve - Invent - Make up - Maximize - Minimize - Modify - Original - Originate - Plan - Predict - Propose - Solution - Solve - Suppose - Test - Theory

KNOWLEDGE INSTITUTE OF TECHNOLOGY(AUTONOMOUS), SALEM – 637504												
M.E. SOFTWARE ENGINEERING										Version : 1.0		
Courses of Study and Scheme of Assessment (Regulation 2023)										Date : 15.12.2025		
SI. No.	Course Code	Course Title	Periods / Week							Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total	
SEMESTER I												
	THEORY											
1.	ME23MA105	Mathematical Modeling and Simulation	FC	4	3	1	0	4	40	60	100	
2.	UX23RM201	Research Methodology and IPR	RM	3	2	1	0	3	40	60	100	
3.	ME23SE301	Advanced Data Structures and Algorithms	PC	3	3	0	0	3	40	60	100	
4.	ME23SE302	Database Design and Technologies	PC	3	3	0	0	3	40	60	100	
5.	ME23SE303	Advanced Software Engineering	PC	3	3	0	0	3	40	60	100	
6.	ME23SE304	Software Architecture	PC	3	3	0	0	3	40	60	100	
	PRACTICALS											
7.	ME23SE305	Software Requirements and Development Laboratory	PC	4	0	0	4	2	60	40	100	
	EMPLOYABILITY ENHANCEMENT											
8.	UX23PT801	Technical Seminar / Case Study Presentation	EEC	2	0	0	2	NC	100	-	100	
TOTAL				25	17	2	6	21	400	400	800	
SEMESTER II												
	THEORY											
1.	ME23SE306	Software Requirements Engineering	PC	3	3	0	0	3	40	60	100	
2.	ME23SE307	Software System Design	PC	3	3	0	0	3	40	60	100	
3.	ME23SE308	Software Testing and Quality Assurance	PC	3	3	0	0	3	40	60	100	
4.	ME23XX5XX	Open Elective – I	OE	3	3	0	0	3	40	60	100	
5.	UX23MC701	Universal Human Values and Ethics	MC	3	1	0	2	NC	100	-	100	
6.	UX23AC7XX	Audit Course	AC	2	2	0	0	NC	100	-	100	
	THEORY CUM PRACTICAL											
7.	ME23SE4XX	Professional Elective - I	PE	5	3	0	2	4	50	50	100	
8.	ME23SE4XX	Professional Elective - II	PE	5	3	0	2	4	50	50	100	
	EMPLOYABILITY ENHANCEMENT											
9.	UX23PT802	Research Paper Review and Presentation	EEC	2	0	0	2	1	100	-	100	
TOTAL				29	21	0	8	21	560	340	900	

w.e.f. 22/12/2025
 Passed in BoS Meeting held on 09/12/2025
 Approved in Academic Council Meeting held on 15/12/2025

KNOWLEDGE INSTITUTE OF TECHNOLOGY(AUTONOMOUS), SALEM – 637504												
M.E. SOFTWARE ENGINEERING												
Courses of Study and Scheme of Assessment (Regulation 2023)												
Sl. No.	Course Code	Course Title	Periods / Week						Maximum Marks			
			CAT	CP	L	T	P	C	IA	ESE	Total	
SEMESTER III												
THEORY												
1.	ME23SE309	Software Agents	PC	3	3	0	0	3	40	60	100	
2.	ME23XX5XX	Open Elective – II	OE	3	3	0	0	3	40	60	100	
THEORY CUM PRACTICAL												
3.	ME23SE4XX	Professional Elective – III	PE	5	3	0	2	4	50	50	100	
4.	ME23SE4XX	Professional Elective – IV	PE	5	3	0	2	4	50	50	100	
PRACTICAL												
5.	ME23SE601	Project Work – Phase I	PW	12	0	0	12	6	60	40	100	
TOTAL				28	12	0	16	20	240	260	500	
SEMESTER IV												
PRACTICAL												
1.	ME23SE602	Project Work –Phase II	PW	24	0	0	24	12	60	40	100	
TOTAL				24	0	0	24	12	60	40	100	
Total Number of Credits: 74												

SEMESTER-WISE CREDITS DISTRIBUTION

SUMMARY							
Sl. No.	Course Category	Credits per Semester				Credits	Credit %
		I	II	III	IV		
1.	FC	4	-	-	-	4	5.4
2.	RM	3	-	-	-	3	4.1
3.	PC	14	9	3	-	26	35.1
4.	PE	-	8	8	-	16	21.6
5.	OE	-	3	3	-	6	8.1
6.	PW	-	-	6	12	18	24.3
7.	MC/AC	✓	✓	-	-	-	-
8.	EEC	-	1	-	-	1	1.4
Total		21	21	20	12	74	100

CAT	Category of Course	FC	Foundation Courses	MC	Mandatory Courses
CP	Contact Period	PC	Professional Core Courses	AC	Audit Courses
L	Lecture Period	PE	Professional Elective Courses	IA	Internal Assessment
T	Tutorial Period	OE	Open Elective Courses	ESE	End Semester Examination
P	Laboratory Period	PW	Project Work Courses		
C	Credits	EEC	Employability Enhancement Courses		

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PROFESSIONAL ELECTIVES

SEMESTER II
(Professional Electives – I & II)

Sl. No.	Course Code	Course Title	Periods / Week						Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
1.	ME23SE401	Frontend Development	PE	5	3	0	2	4	50	50	100
2.	ME23SE402	Backend Development with Java	PE	5	3	0	2	4	50	50	100
3.	ME23SE403	Full stack Development and Deployment	PE	5	3	0	2	4	50	50	100
4.	ME23SE404	Fundamentals of Web Development	PE	5	3	0	2	4	50	50	100
5.	ME23SE405	Testing Fundamentals and Manual Testing	PE	5	3	0	2	4	50	50	100
6.	ME23SE406	Automation Testing with Selenium	PE	5	3	0	2	4	50	50	100

SEMESTER III
(Professional Electives – III & IV)

7.	ME23SE407	Devops and Cloud Basics	PE	5	3	0	2	4	50	50	100
8.	ME23SE408	Agile based Project Automation with Devops	PE	5	3	0	2	4	50	50	100
9.	ME23SE409	CI and CD(Continuous Integration and Continuous Development)	PE	5	3	0	2	4	50	50	100
10.	ME23SE410	Advanced Testing with TestNG and CI/CD	PE	5	3	0	2	4	50	50	100
11.	ME23SE411	Tools for Automated Testing	PE	5	3	0	2	4	50	50	100
12.	ME23SE412	Performance and Database Testing	PE	5	3	0	2	4	50	50	100

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OPEN ELECTIVES I & II												
Sl. No.	Course Code	Course Title	Offered by	Periods / Week						Maximum Marks		
				CAT	CP	L	T	P	C	IA	ESE	Total
Except M.E. Industrial Safety Engineering												
1.	ME23IS501	Environmental Safety	MECH	OE	3	3	0	0	3	40	60	100
2.	ME23IS502	Electrical Safety	MECH	OE	3	3	0	0	3	40	60	100
3.	ME23IS503	Safety in Engineering Industry	MECH	OE	3	3	0	0	3	40	60	100
4.	ME23IS504	Design of Experiments	MECH	OE	3	3	0	0	3	40	60	100
5.	ME23IS505	Circular Economy	MECH	OE	3	3	0	0	3	40	60	100
Except M.E. Automotive Electronics												
6.	ME23AE501	Fundamentals of Automotive Electronics	ECE	OE	3	3	0	0	3	40	60	100
7.	ME23AE502	Vehicle Sensors and Electronic Control Systems	ECE	OE	3	3	0	0	3	40	60	100
8.	ME23AE503	Automotive Embedded Systems	ECE	OE	3	3	0	0	3	40	60	100
9.	ME23AE504	Automotive Communication Networks	ECE	OE	3	3	0	0	3	40	60	100
10.	ME23AE505	Introduction to Electric and Hybrid Vehicle Electronics	ECE	OE	3	3	0	0	3	40	60	100
Except M.E. Power Electronics and Drives												
11.	ME23PE501	Electric Vehicle Technology	EEE	OE	3	3	0	0	3	40	60	100
12.	ME23PE502	Energy Conservation and Management	EEE	OE	3	3	0	0	3	40	60	100
13.	ME23PE503	Smart Grid	EEE	OE	3	3	0	0	3	40	60	100
14.	ME23PE504	Solar PV and Energy Storage Systems	EEE	OE	3	3	0	0	3	40	60	100
Except M.E. Software Engineering												
15.	ME23SE501	Blockchain Technologies	CSE	OE	3	3	0	0	3	40	60	100
16.	ME23SE502	Deep Learning	CSE	OE	3	3	0	0	3	40	60	100
17.	ME23SE503	Full Stack Web Application Development	CSE	OE	3	3	0	0	3	40	60	100
18.	ME23SE504	Principles of Multimedia	CSE	OE	3	3	0	0	3	40	60	100
19.	ME23SE505	Cyber Physical Systems	CSE	OE	3	3	0	0	3	40	60	100
AUDIT COURSES												
1.	UX23AC701	Industrial Safety Engineering	MECH	AC	2	2	0	0	0	100	-	100
2.	UX23AC702	Sustainability Engineering	MECH	AC	2	2	0	0	0	100	-	100
3.	UX23AC703	Energy Audit	EEE	AC	2	2	0	0	0	100	-	100
4.	UX23AC704	Design Thinking	CSE	AC	2	2	0	0	0	100	-	100
5.	UX23AC705	Business Analytics	CSE	AC	2	2	0	0	0	100	-	100

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ME23MA105	MATHEMATICAL MODELING AND SIMULATION	CP	L	T	P	C
Programme & Branch	M.E- SOFTWARE ENGINEERING	4	3	1	0	4
		Version: 1.0				
Course Objectives:						
1.	To Acquire the knowledge of solving system of linear equations using an appropriate numerical methods.					
2.	To approximate the functions using polynomial interpolation numerical differentiation and integration using interpolating polynomials.					
3.	To acquire the knowledge of numerical solution of ordinary differential equation by single and multi step methods.					
4.	To obtain the solution of boundary value problems in partial differential equations using finite differences.					
5.	To study simulation and Monte-Carlo methods and their applications.					
UNIT-I	MATRICES AND LINEAR SYSTEMS OF EQUATIONS					9+3
	Solution of Linear Systems : Cramer's Rule - Gaussian elimination and Gauss Jordan methods - Cholesky decomposition method - Gauss Seidel iteration method - Eigenvalue problems: Power method with deflation for symmetric matrices .					
UNIT-II	INTERPOLATION, DIFFERENTIATION AND INTEGRATION					9+3
	Lagrange's Interpolation - Newton's divided differences - Hermite's Interpolation - Newton's forward and backward differences - Numerical differentiation - Numerical Integration : Trapezoidal and Simpson's 1/3 rules - Gaussian quadrature - 2 and 3 point rules .					
UNIT-III	DIFFERENTIAL EQUATIONS					9+3
	Initial value problems for first and second order ODEs : Single step methods - Taylor's series method - Euler's and modified Euler's methods - Runge - Kutta method of fourth order - Multi step methods : Milne's and Adam Bashforth methods - Boundary value problems : Finite difference approximations to derivatives.					
UNIT - IV	PARTIAL DIFFERENTIAL EQUATIONS					9+3
	Classification of second order PDE's - Finite difference approximations to partial derivatives - Elliptic equations : Solution of Laplace and Poisson equations - One dimensional parabolic equation - Bender Schmidt method - Hyperbolic equation : One dimensional wave equation.					
UNIT-V	SIMULATION AND MONTE CARLO METHODS					9+3
	Random numbers : Random number algorithms and generators - Estimation of areas and volumes by Monte Carlo techniques - Simulation : Loaded Die Problem - Birthday problem - Buffon's needle problem.					
						Total (LT): 60 Periods

w.e.f. 13/10/2025

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8

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Develop proficiency in solving systems of linear equations using appropriate numerical methods.	L3 – Apply
CO2	Apply polynomial interpolation techniques for function approximation, and perform numerical differentiation and integration using interpolating polynomials.	L3 – Apply
CO3	Gain expertise in solving ordinary differential equations (ODEs) using both single-step and multi-step numerical methods	L3 – Apply
CO4	Solve boundary value problems in partial differential equations through finite difference methods.	L3 – Apply
CO5	Apply simulation and Monte Carlo methods for solving real-world problems.	L3 – Apply
REFERENCE BOOKS:		
1.	Maki, D P and Thompson, M., "Mathematical Modelling with Computer Simulation", Cengage Learning, Delhi, 2011.	
2.	Jain, M.K., Iyengar, S.R.K. and Jain R.K. "Numerical Methods for Scientific and Engineering Computation", 6th Edition, New Age International Pvt. Ltd., Delhi, 2014.	
3.	Cheney, W and Kincaid D. "Numerical Mathematics and Computing", 7th Edition, Cengage Learning, Delhi, 2014.	

Mapping of COs with POs						
COs	POs					
	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3			
CO2	2		3			
CO3	2		3			
CO4	2		3			
CO5	2		3			
Average	2.0		3.0			

1-Low, 2 -Medium, 3-High.

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UX23RM201	RESEARCH METHODOLOGY AND IPR	CP	L	T	P	C
		3	2	1	0	3
Programme & Branch	Common to all M.E. (SE,AE,PED,ISE) and M.C.A.	Version: 1.0				
Course Objectives:						
1.	To analyze the significance of research and formulate well-defined research questions.					
2.	To apply appropriate research methods and critically evaluate research articles.					
3.	To create well-structured research papers and utilize research tools proficiently.					
4.	To produce effective technical reports and deliver impactful presentations.					
5.	To understand forms of intellectual property and analyze their implications on technological research and international cooperation.					
UNIT-I	CONCEPT OF RESEARCH					6+3
	Meaning and Significance of Research - Skills, Habits and Attitudes for Research - Time Management - Status of Research in India - Why, How, and What a Research is? - Types and Process of Research - Outcome of Research - Sources of Research Problem - Characteristics of a Good Research Problem - Errors in Selecting a Research Problem - Importance of Keywords - Literature Collection - Analysis - Citation Study - Gap Analysis - Problem Formulation Techniques.					
UNIT-II	RESEARCH METHODS AND JOURNALS					6+3
	Interdisciplinary Research - Need for Experimental Investigations - Data Collection Methods - Appropriate Choice of Algorithms Methodologies / Methods - Measurement and Result Analysis - Investigation of Solutions for Research Problem - Interpretation - Research Limitations - Journals in Science/Engineering - Indexing and Impact factor of Journals - Citations - h Index - i10 Index - Journal Policies - How to Read a Published Paper - Ethical Issues Related to Publishing - Plagiarism and Self-Plagiarism .					
UNIT-III	PAPER WRITING AND RESEARCH TOOLS					6+3
	Types of Research Papers - Original Article/Review Paper/Short Communication/Case Study - When and Where to Publish? - Journal Selection Methods - Layout of a Research Paper - Guidelines for Submitting the Research Paper - Review Process - Addressing Reviewer Comments - Use of tools / Techniques for Research - Hands - on Training related to Reference Management Software - EndNote - Introduction to Origin, SPSS, etc - Software for Detection of Plagiarism.					
UNIT-IV	EFFECTIVE TECHNICAL THESIS WRITING/PRESENTATION					6+3
	How to Write a Report - Language and Style - Format of Project Report - Use of Quotations - Method of Transcription Special					

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10


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	Elements - Title Page - Abstract - Table of Contents - Headings and Sub-Headings - Footnotes - Tables and Figures - Appendix - Bibliography etc. - Different Reference Formats - Presentation using PPTs .	
UNIT-V	NATURE OF INTELLECTUAL PROPERTY	6+3
	Patents - Designs - Trade and Copyright - Process of Patenting and Development - Technological research - Innovation - Patenting - Development International Scenario - International Cooperation on Intellectual Property - Procedure for Grants of Patents.	
	Total (LT)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Bloom's Taxonomy Levels: Remember(L1), Understand (L2), Apply (L3), Analyze (L4), Evaluate (L5) and Create (L6)	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Illustrate the importance and objectives of research in contributing to knowledge and solving real-world problems.	L2 - Understand
CO2	Experiment with data collection techniques, choosing fitting approaches to ensure sound research framework and methodology.	L3 - Apply
CO3	Interpret the components and structure of research papers and apply this knowledge to create organized and effective academic documents.	L2 - Understand
CO4	Apply knowledge to produce engaging presentations and detailed technical reports that effectively communicate research findings.	L3 - Apply
CO5	Differentiate between types of intellectual property and comprehend	L4 - Analyze
	REFERENCE BOOKS:	
1.	Cooper Donald R, Schindler Pamela S and Sharma JK, "Business Research Methods", 11e , TataMcGraw Hill Education, 2012.	
2.	DePoy, Elizabeth, and Laura N. Gitlin, "Introduction to Research-E-Book: Understanding and Applying Multiple Strategies", Elsevier Health Sciences, 2015.	
3.	Walliman, Nicholas, "Research Methods: The basics", Routledge, 2017.	
4.	Bettig Ronald V., "Copyrighting culture: The political economy of intellectual property", Routledge, 2018.	

w.e.f. 13/10/2025
 Passed in BoS Meeting held on 10/05/2025
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Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	3		1		2
CO2	3	3		1		2
CO3	3	3		1		2
CO4	3	3		1		2
CO5	3	3				2
Average	3.0	3.0		1.0		2.0
1-Low, 2 -Medium, 3-High						




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ME23SE301	ADVANCED DATA STRUCTURES AND ALGORITHMS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E- SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand the usage of algorithms in computing.					
2.	To learn and use hierarchical data structures and its operations.					
3.	To learn the usage of graphs and its applications.					
4.	To select and design data structures and algorithms that is appropriate for problems.					
5.	To study about NP Completeness of problems.					
UNIT-I	ROLE OF ALGORITHMS IN COMPUTING & COMPLEXITY ANALYSIS					9
	Algorithms - Algorithms as a Technology - Time and Space complexity of algorithms - Asymptotic analysis -Average and worst-case analysis - Asymptotic notation - Importance of efficient algorithms - Program performance measurement - Recurrences: The Substitution Method - The Recursion-Tree Method - Data structures and algorithms.					
UNIT-II	HIERARCHICAL DATA STRUCTURES					9
	Binary Search Trees: Basics - Querying a Binary search tree - Insertion and Deletion - Red Black trees: Properties of Red-Black Trees - Rotations - Insertion - Deletion - B - Trees: Definition of B - trees - Basic operations on B-Trees - Deleting a key from a B - Tree - Heap - Heap Implementation - Disjoint Sets - Fibonacci Heaps: structure - Mergeable - heap operations - Decreasing a key and deleting a node - Bounding the maximum degree.					
UNIT-III	GRAPHS					9
	Elementary Graph Algorithms: Representations of Graphs - Breadth-First Search - Depth- First Search - Topological Sort - Strongly Connected Components - Minimum Spanning Trees: Growing a Minimum Spanning Tree - Kruskal and Prim - Single-Source Shortest Paths: The Bellman-Ford algorithm - Single-Source Shortest paths in Directed Acyclic Graphs - Dijkstr's Algorithm, Dynamic Programming - All-Pairs Shortest Paths: Shortest Paths and Matrix Multiplication - The Floyd-Warshall Algorithm.					
UNIT-IV	ALGORITHM DESIGN TECHNIQUES					9
	Dynamic Programming: Matrix-Chain Multiplication - Elements of Dynamic Programming - Longest Common Subsequence - Greedy					

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	Algorithms: - Elements of the Greedy Strategy - An Activity - Selection Problem - Huffman Coding.	
UNIT-V	NP COMPLETE AND NP HARD	9
	NP-Completeness: Polynomial Time – Polynomial-Time Verification – NP - Completeness and Reducibility – NP-Completeness Proofs – NP-Complete Problems.	
	Total (L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Bloom's Taxonomy Levels: Remember(L1), Understand (L2), Apply (L3), Analyze (L4), Evaluate (L5) and Create (L6)	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Illustrate the time and space complexity of algorithms and perform asymptotic analysis.	L3 – Apply
CO2	Demonstrate the use of efficient data structures for practical problem-solving.	L3 – Apply
CO3	Determine graph structures and string-matching algorithms to design solutions for real-life problems.	L3 – Apply
CO4	Develop solutions for computational problems by selecting and applying appropriate dynamic programming or greedy approaches.	L3 – Apply
CO5	Apply the principles of NP-completeness and reducibility to classify computational problems.	L3 – Apply
	REFERENCE BOOKS:	
1.	S. Sridhar, "Design and Analysis of Algorithms", 1 st Edition, Oxford University Press, 2014.	
2.	Adam Drozdex, "Data Structures and algorithms in C++", 4 th Edition, Cengage Learning, 2013.	
3.	T.H. Cormen, C.E.Leiserson, R.L. Rivest and C.Stein, "Introduction to Algorithms", 3 rd Edition, PrenticeHall of India, 2012.	
4.	Mark Allen Weiss, "Data Structures and Algorithms in C++", 3 rd Edition, Pearson Education, 2009.	
5.	E. Horowitz, S. Sahni and S. Rajasekaran, "Fundamentals of Computer Algorithms", 2 nd Edition, University Press, 2008.	

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Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		1	2		1
CO2	3		1	2		1
CO3	2		1	2		1
CO4	2		1	2		1
CO5	2		1	2		1
Average	2.4		1.0	2.0		1.0
1-Low, 2 -Medium, 3-High						




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ME23SE302	DATABASE DESIGN AND TECHNOLOGIES	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E- SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To apply relational algebra operations and SQL commands for data retrieval and manipulation.					
2.	To examine the techniques used to improve query performance in distributed environments.					
3.	To construct well-formed XML documents and validate them using DTD and XML Schema.					
4.	To classify the different types of NoSQL databases (key-value, column, document, graph).					
5.	To identify different database security models.					
UNIT-I	RELATIONAL DATA MODEL					9
	Entity Relationship Model – Relational Data Model – Mapping Entity Relationship Model to Relational Model – Relational Algebra – Structured Query Language – Database Normalization.					
UNIT-II	DISTRIBUTED DATABASES, ACTIVE DATABASES AND OPEN DATABASE CONNECTIVITY					9
	Distributed Database Architecture – Distributed Data Storage – Distributed Transactions – Distributed Query Processing – Distributed Transaction Management – Event Condition Action Model – Design and Implementation Issues for Active Databases – Open Database Connectivity.					
UNIT-III	XML DATABASES					9
	Structured, Semi structured, and Unstructured Data – XML Hierarchical Data Model – XML Documents – Document Type Definition – XML Schema – XML Documents and Databases XML Querying – XPath – XQuery.					
UNIT-IV	NOSQL DATABASES AND BIG DATA STORAGE SYSTEMS					9
	NoSQL – Categories of NoSQL Systems – CAP Theorem – Document-Based NoSQL Systems and MongoDB – MongoDB Data Model – MongoDB Distributed Systems Characteristics – NoSQL Key-Value Stores – DynamoDB Overview – Voldemort Key-Value Distributed Data Store – Wide Column NoSQL Systems – Hbase Data Model – Hbase Crud Operations – Hbase Storage and Distributed System Concepts – NoSQL Graph Databases and Neo4j – Cypher Query Language of Neo4j – Big Data – MapReduce – Hadoop – YARN.					

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UNIT-V	DATABASE SECURITY	9
	Database Security Issues – Discretionary Access Control Based on Granting and Revoking Privileges – Mandatory Access Control and Role-Based Access Control for Multilevel Security – SQL Injection – Statistical Database Security – Flow Control – Encryption and Public Key Infrastructures – Preserving Data Privacy – Challenges to Maintaining Database Security – Database Survivability – Oracle Label-Based Security.	
	Total (L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Bloom's Taxonomy Levels: Remember(L1), Understand (L2), Apply (L3), Analyze (L4), Evaluate (L5) and Create (L6)	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Determine relational tables from an ER-model and populate the database to support SQL queries.	L3 - Apply
CO2	Apply techniques for distributed query processing to optimize database performance.	L3 - Apply
CO3	Illustrate the creation of well-formed XML documents and validate their structure.	L3 - Apply
CO4	Experiment with data definition, manipulation, and control commands in NoSQL databases to manage and query data effectively.	L3 - Apply
CO5	Model secure database designs to implement systems that protect data integrity and confidentiality.	L3 - Apply
	REFERENCE BOOKS:	
1.	R. Elmasri, S.B. Navathe, "Fundamentals of Database Systems", Seventh Edition, Pearson Education 2016.	
2.	Henry F Korth, Abraham Silberschatz, S. Sudharshan, "Database System Concepts", Seventh Edition, McGraw Hill, 2019.	
3.	C.J.Date, A.Kannan, S.Swamynathan, "An Introduction to Database Systems, Eighth Edition, Pearson Education, 2006.	
4.	Raghu Ramakrishnan, Johannes Gehrke "Database Management Systems", Fourth Edition, McGraw Hill Education, 2015.	
5.	Harrison, Guy, "Next Generation Databases, NoSQL and Big Data", First Edition, Apress publishers, 2015.	
6.	Thomas Cannolly and Carolyn Begg, "Database Systems, A Practical Approach to Design, Implementation and Management", Sixth Edition, Pearson Education, 2015.	

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Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
C01	3		2	2		1
C02	3		2	2		1
C03	3		2	2		1
C04	3		2	2		1
C05	2		2	1		1
Average	2.8		2.0	1.8		1.0
1-Low, 2 -Medium, 3-High						




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ME23SE303	ADVANCED SOFTWARE ENGINEERING	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E- SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand the rationale for software development process models.					
2.	To gain the knowledge of architectural design of software.					
3.	To understand the five important dimensions of dependability, namely, availability, reliability, safety, security, and resilience.					
4.	To explore the basic notions of web service, web service standards, and service-oriented architecture.					
5.	To understand the different stages of testing during development of a software system.					
UNIT-I	SOFTWARE PROCESS &MODELING	9				
	Prescriptive Process Models - Agility and Process - Scrum - XP - Kanban - DevOps - Prototype Construction - Prototype Evaluation - Prototype Evolution - Modeling - Principles - Requirements Engineering - Scenario-based Modeling - Class-based Modeling - Functional Modeling - Behavioral Modeling.					
UNIT-II	SOFTWARE DESIGN	9				
	Design Concepts - Design Model - Software Architecture - Architectural Styles - Architectural Design - Component-Level Design - User Experience Design - Design for Mobility - Pattern Based Design.					
UNIT-III	SYSTEM DEPENDABILITY AND SECURITY	9				
	Dependable Systems - Dependability Properties - Sociotechnical Systems - Redundancy and Diversity - Dependable Processes - Formal Methods and Dependability - Reliability Engineering - Availability and Reliability - Reliability Requirements - Fault-tolerant Architectures - Programming for Reliability - Reliability Measurement - Safety Engineering - Safety-critical Systems - Safety Requirements - Safety Engineering Processes - Safety Cases - Security Engineering - Security and Dependability - Safety and Organizations - Security Requirements - Secure System Design - Security Testing and Assurance - Resilience Engineering - Cybersecurity - Sociotechnical Resilience - Resilient Systems Design.					
UNIT-IV	SERVICE-ORIENTED SOFTWARE ENGINEERING AND REAL-TIME SOFTWARE ENGINEERING	9				
	Service-oriented Architecture - RESTful Services - Service Engineering - Service Composition - Systems Engineering - Sociotechnical Systems - Conceptual Design - System					

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	Procurement – System Development – System Operation and Evolution – Real-time Software Engineering – Embedded System Design – Architectural Patterns for Real-time Software – Timing Analysis – Real-time Operating Systems.	
UNIT-V	SOFTWARE TESTING AND PROJECT MANAGEMENT	9
	Software Testing Strategy - Project management and costing - Process models and lifecycle management - Risk management - Metrics-people management - Software configuration management.	
	Total (L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Bloom's Taxonomy Levels: Remember(L1), Understand (L2), Apply (L3), Analyze (L4), Evaluate (L5) and Create (L6)	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Identify appropriate process models based on the Project requirements.	L2 – Understand
CO2	Understand the importance of having a good Software Architecture.	L2 – Understand
CO3	Describe the five important dimensions of dependability namely, availability, reliability, safety, security, and resilience.	L2 – Understand
CO4	Understand the basic notions of a web service, web service standards, and service-oriented architecture.	L2 – Understand
CO5	Interpret various levels of Software testing.	L3 – Apply
	REFERENCE BOOKS:	
1.	Roger Pressman and Bruce Maxim, "Software Engineering : A Practitioner's Approach", 9 th Edition, McGraw-Hill 2019.	
2.	Software Engineering, 10 th Edition, Ian Somerville, Pearson Education Asia 2016.	
3.	Len Bass, Paul Clements and Rick Kazman, "Software Architecture in Practice", 3 rd Edition, Pearson India, 2018.	
4.	Pankaj Jalote, "An integrated approach to Software Engineering", 3 rd Edition, Narosa Publishing House, 2018.	
5.	Rajib Mall, "Fundamentals of Software Engineering", 5 th Edition, PHI Learning Private Ltd, 2018.	

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Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1		3	3	2	2
CO2	1		3	3	2	2
CO3	1		3	3	2	2
CO4	1		3	3	2	2
CO5	1		3	3	3	1
Average	1.0		3.0	3.0	2.2	1.8
1-Low, 2 -Medium, 3-High						




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ME23SE304	SOFTWARE ARCHITECTURE	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E- SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand the fundamentals of software architecture.					
2.	To study the various software modeling techniques.					
3.	To understand software implementation and deployment.					
4.	To learn the architecture of different applications.					
5.	To relate software architecture and software quality.					
UNIT-I	BASIC CONCEPTS, DESIGNING ARCHITECTURES					9
	Understanding Software Architecture-Concepts of Software Architecture, Models, Processes, Stakeholders, the Design Process, Architectural Conception, Refined Experience in Action: Styles and Architectural Patterns, Architectural Conception in Absence of Experience. Suggested Activities: 1. Identifying the pitfalls that are likely to occur for software architecture teams 2. Discussing about the role of Software architects as Cost estimators.					
UNIT-II	CONNECTORS, MODELING					9
	Connectors in Action: A Motivating Example, Connector Foundations, Connector Roles, Connector Types and Their Variation Dimensions, Example Connectors, Modeling Concepts, Ambiguity, Accuracy, and Precision, Complex Modeling: Mixed Content and Multiple Views, Evaluating Modeling Techniques, Specific Modeling Techniques. Suggested Activities: 1. Identifying a few commercial products which are infrastructure components that provide asynchronous messaging service. 2. Finding out the different possibilities of splitting the system into a number of computationally independent execution structures.					
UNIT-III	ANALYSIS, IMPLEMENTATION AND DEPLOYMENT					9
	Concepts, Existing Frameworks, Software Architecture and Deployment, Software Architecture and Mobility, Pipes and Filters, Event- based, Implicit Invocation, Layered systems, Repositories Interpreters, Process control.					

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	Suggested Activities: 1. Identifying the type of a given architectural pattern 2. Representing software using pipe-filter architecture.	
UNIT-IV	APPLIED ARCHITECTURES AND STYLES	9
	Distributed and Networked Architectures, Architectures for Network-Based Applications, Decentralized Architectures, Service-Oriented Architectures and Web Services, Efficiency, Complexity, Scalability and Heterogeneity, Adaptability, Dependability. Suggested Activities: 1. Identifying functional aspects of a service oriented architecture. 2. Discussing the pros and cons of implementing a middle ware to deal with architectural issues.	
UNIT-V	IMPLEMENTATION	9
	Understanding quality attributes- Availability- Deployability - Working with Other Quality Attributes-Virtualization - the Cloud and Distributed Computing - Architecturally Significant Requirements- Designing an Architecture. Suggested Activity: 1. Identifying the cost of modifications in projects that measure deployment separately. 2. Using the cloud assumes your application is service oriented. Find some examples that would support that statement and, if it is not universally true, find some that would falsify it.	
	Total (L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Bloom's Taxonomy Levels: Remember(L1), Understand (L2), Apply (L3), Analyze (L4), Evaluate (L5) and Create (L6)	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Examine software architecture concepts and assess the role of software architects in identifying pitfalls and estimating costs.	L3 - Apply
CO2	Apply different types of systems analysis techniques and software design strategies.	L3 - Apply
CO3	Implement architectural styles like pipe-filter, event-driven, and layered systems to model software behavior.	L3 - Apply
CO4	Apply suitable design patterns based on system requirements and functionality.	L3 - Apply

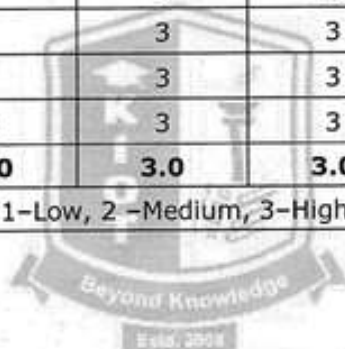
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CO5	Demonstrate the quality attributes for software architecture.	L3 – Apply
REFERENCE BOOKS:		
1.	Richard N, Taylor, Nenad Medvidovic, Eric Dashofy, "Software Architecture: Foundations, Theory, and Practice", 2009.	
2.	Len Bass, Paul Clements, Rick Kazman, "Software Architecture in Practice", 4 th Edition, Pearson, 2021.	
3.	M. Shaw and D Garlan, "Software Architecture Perspectives on an Emerging Discipline", Prentice- Hall, Digitized version 2007.	
4.	Mark Richards, Neal Ford, "Fundamentals of Software Architecture, An Engineering Approach", O'Reilly Media, 2020.	

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1		3	3	2	2
CO2	1		3	3	2	2
CO3	1		3	3	2	2
CO4	1		3	3	2	2
CO5	1	1	3	3	3	1
Average	1.0	1.0	3.0	3.0	2.2	1.8
1-Low, 2 -Medium, 3-High						




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ME23SE305	SOFTWARE REQUIREMENTS AND DEVELOPMENT LABORATORY	CP	L	T	P	C
Programme & Branch	M.E- SOFTWARE ENGINEERING	4	0	0	4	2
		Version: 1.0				
Course Objectives:						
1.	To apply project management methods to study problems and determine scope, goals, and resources.					
2.	To apply structured techniques to collect requirements and prepare a Software Requirements Specification					
3.	To apply design methods to develop software architecture and model system behavior using UML diagrams.					
4.	To apply coding practices and standards to build a working web application.					
5.	To apply testing and deployment procedures to validate functionality and deliver an installable system.					
LIST OF EXPERIMENTS/EXERCISES:						
Choose any one application for performing the following phases						
1.	Program Analysis and Project Planning - Thorough study of the problem - Identify project scope, Objectives, Infrastructure.					
2.	Software Requirements- Analysis using a structured approach and producing SRS Artifact.					
3.	Software Design and Data Modeling - producing Software Architecture, Use case diagrams, Class Diagrams and activity diagrams, Sequence diagrams, Deployment Diagram.					
4.	Software Implementation - Creating a simple web application of any problem statement consider at the beginning phase, following with proper coding conventions and standard.					
5.	Software Testing - Perform unit testing and system testing using any testing framework.					
6.	Software Deployment - Perform deployment builds and create jars with complement environment for installation.					
SUGGESTED LIST OF APPLICATIONS:						
1.	Inventory System.					
2.	Book Lending Application.					
3.	Car Booking Application.					
4.	Online Help Portal.					
5.	Student Marks Analyzing System.					
6.	Credit Card Processing.					
7.	Stock Maintenance System.					

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8.	Airline Reservation System.	
9.	Recruitment system.	
10.	Course Registration System.	
	Total (P)	60 Periods
HARDWARE/SOFTWARE REQUIREMENTS		
1.	64-bit Open source Linux/windows or its derivative.	
2.	IBM rational suite/Microsoft visual studio enterprise / GitLab.	
	Bloom's Taxonomy Levels: Remember(L1), Understand (L2), Apply (L3), Analyze (L4), Evaluate (L5) and Create (L6)	
Course Outcomes: Upon completion of this course, the students will be able to:		BLOOM'S Taxonomy
CO1	Apply project management principles to define project scope, objectives, and infrastructure.	L3 – Apply
CO2	Apply structured analysis to gather requirements and create an SRS document.	L3 – Apply
CO3	Apply design principles to produce architecture and UML diagrams for the system.	L3 – Apply
CO4	Apply coding standards to develop a functional web application.	L3 – Apply
CO5	Apply testing and deployment techniques to ensure a working and installable system.	L3 – Apply

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1	1	1	1		3
CO2	2	1	1	2		3
CO3	2	3	1	2		3
CO4	2	3	1			3
CO5	2	3	1	1	3	3
Average	1.8	2.2	1	1.5	3.0	3.0
1-Low, 2 –Medium, 3-High						

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UX23PT801	TECHNICAL SEMINAR / CASE STUDY PRESENTATION	CP	L	T	P	C
		2	0	0	2	0
Programme & Branch	Common to M.E. (SE,AE,PED,ISE)	Version: 1.0				

Course Objectives:

1.	To encourage the students to study advanced engineering developments.
2.	To prepare and present the technical and case study reports.

Method of Evaluation:

The students need to identify an area of interest or topic in their programme of study or case study and prepare a 5-10 page report and a presentation. Based on the report and presentation, the course is evaluated for 100 marks. Minimum 50 marks is essential to pass. In case a student fails, he / she has to make such presentation in the subsequent semesters. The evaluation guidelines will be issued by the Head of the Department before the commencements of the course. The objectives are improving literature searching capabilities, comprehension and ability to write reports and to make presentations. It is assessed in Internal Assessment mode only and no End Semester Examination.

	Total (P)	30 Periods
	Bloom's Taxonomy Levels: Remember(L1), Understand (L2), Apply (L3), Analyze (L4), Evaluate (L5) and Create (L6)	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Perform the review and present technological developments in their field	L3 – Apply
CO2	Interpret the case study report and make a decision	L3 – Apply

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1	3		3		1
CO2	1	3		3		1
Average	1	3		3		1

1–Low, 2 –Medium, 3–High

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ME23SE306	SOFTWARE REQUIREMENTS ENGINEERING	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E- SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand the basics of requirements engineering.					
2.	To learn different techniques used for requirements elicitation.					
3.	To know the role played by requirements analysis in requirement integration.					
4.	To appreciate the use of various methodologies for requirements development.					
5.	To study the current trends in requirements prioritization and validation.					
UNIT-I	INTRODUCTION	9				
	Introduction to Requirements - System - Stakeholder-Requirement - Attribute - Vision- Function - Performance - Objective - Quality- Resource Saving - Workload Capacity - Resource- Cost - Budget Design Idea - Condition - Target - Constraint - Benchmark - Introduction to Requirements Engineering - Requirements Engineering activities - Understanding requirements - classification based on functionality 24 considerations, product construction, source-Levels of requirement-Evolution -Ambiguity in requirements specification.					
UNIT-II	REQUIREMENTS ELICITATION	9				
	Requirements Elicitation: Preparing for requirements elicitation. Elicitation techniques surveyeliciting non - functional requirements - prevalence of requirements elicitation techniques - eliciting hazards - Elicitation and gathering of requirements - Process of requirements elicitation - approaches for requirements elicitation methodology based requirements elicitation - tool support - issues - trends and challenges.					
UNIT-III	REQUIREMENTS ANALYSIS	9				
	Identification of Functional and Non Functional Requirements - Identification of Performance Requirements - Six Quality Attributes - Analysis - Identification of safety Requirements - Analysis - Feasibility and Internal Compatibility of System Requirements - Definition of Human Requirements Baseline.					
UNIT-IV	REQUIREMENTS DEVELOPMENT	9				
	Requirements analysis - Requirements Documentation - Requirements Development Workflow - Fundamentals of Requirements Development - Requirements Attributes Guidelines					

w.e.f. 22/12/2025

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	Document – Supplementary Specification Document – Use Case Specification Document – Methods for Software Prototyping – Evolutionary prototyping – Throwaway prototyping.	
UNIT-V	REQUIREMENTS VALIDATION	9
	Validation objectives – Analysis of requirements validation – Activities – Properties – Requirement reviews – Requirements testing – Case tools for requirements engineering – Requirements engineering for Agile methods – Requirements Engineering for Web-Based Information Systems.	
	Total (L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Apply the principles of requirements engineering to prepare a detailed Software Requirements Specification.	L3 – Apply
CO2	Implement suitable techniques for effective requirements elicitation.	L3 – Apply
CO3	Analyze software requirements gathering.	L3 – Apply
CO4	Apply various methodologies for requirements development.	L3 – Apply
CO5	Demonstrate the process of validating software requirements.	L3 – Apply
	REFERENCE BOOKS:	
1.	Wieggers, Karl, Joy Beatty, "Software requirements", Pearson Education, 2013.	
2.	Phillip A. Laplante "Requirements Engineering for Software and Systems", Auerbach Publications(Applied Software Engineering Series) - (2017).	
3.	Aybüke Aurum, Claes Wohlin, "Engineering and managing software requirements", Springer-Verlag Berlin Heidelberg, 2005.	
4.	Ian Sommerville, Pete Sawyer, "Requirements engineering: A Good Practice Guide", Sixth Edition, Pearson Education, 2004.	
5.	Dean Leffingwe, Don Widrig, "Managing Software Requirements A Use Case Approach", Second Addition, Addison Wesley, 2003	

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Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3	3		2
CO2	2		3	3		2
CO3	2		3	3		2
CO4	2		3	3		2
CO5			3	2	3	
Average	2.0		3.0	2.8	3.0	2.0
1-Low, 2 -Medium, 3-High						




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ME23SE307	SOFTWARE SYSTEM DESIGN	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E- SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand the fundamentals of object modeling.					
2.	To learn the unified process phases.					
3.	To prepare the requirements for various case studies.					
4.	To appreciate the idea behind Design Patterns in handling common problems faced during building an application.					
5.	To practice object modeling using UML.					
UNIT-I	INTRODUCTION	9				
	Introduction to OOAD - typical activities / workflows / disciplines in OOAD - Introduction to iterative development and the Unified Process, Introduction to UML, mapping disciplines to UML artifacts, Introduction to Design Patterns - goals of a good design, Introducing a case study & MVC architecture.					
UNIT-II	INCEPTION	9				
	Artifacts in inception, Understanding requirements - the FURPS model, Understanding Use case model - introduction - use case types and formats - Writing use cases - goals and scope of a use case - elements / sections of a use case - Use case diagrams - Use cases in the UP context and UP artifacts - Identifying additional requirements - Writing requirements for the case study in the use case model.					
UNIT-III	ELABORATION	9				
	System sequence diagrams for use case model- Domain model : identifying concepts - adding associations - adding attributes - Interaction Diagrams - Introduction to GRASP design Patterns Design Model: Use case realizations with GRASP patterns - Design Class diagrams in each MVC layer Mapping Design to Code - Design class diagrams for case study and skeleton code.					
UNIT-IV	DESIGN PATTERNS	9				
	Fabrication - Indirection - Singleton - Factory - Façade - Publish-Subscribe.					
UNIT-V	UML DIAGRAMS	9				
	State-Chart diagrams - Activity diagrams - Component Diagrams - Deployment diagrams - Object diagrams. Advanced concepts in					

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	OOAD : Use case relationships - Generalizations Domain Model refinements - Architecture - Packaging model elements.	
	Total (L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Apply UML Use case Notations to applications.	L3 – Apply
CO2	Apply unified process in software development.	L3 – Apply
CO3	Demonstrate the best use of Object-Oriented concepts for creating truly OOP programs.	L3 – Apply
CO4	Illustrate design patterns for better class and object composition.	L3 – Apply
CO5	Examine the concepts of Model refinement and diagrams.	L3 – Apply
	REFERENCE BOOKS:	
1.	Craig Larman, "Applying UML and patterns", Pearson, 2005.	
2.	Satzinger, Jackson, Burd "Object-Oriented Analysis & Design with the Unified Process", Cengage Learning India Publisher, 2007.	
3.	Martin Fowler, "UML distilled", Addison Wesley, Third Edition, 2003.	
4.	Eric Freeman et al, "Head-First Design Patterns", O'Reilly's, Second Edition, 2004.	
5.	Hans-Erik Eriksson, Magnus Penker, Brian Lyons, David Fado, "UML2 Toolkit", Wiley, 2003.	

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3	2		
CO2	2		3	3		
CO3	1		3	2		
CO4	2		3	2		
CO5	2		3			
Average	2.0		3.0	2.0		
1-Low, 2 –Medium, 3-High						

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ME23SE308	SOFTWARE TESTING AND QUALITY ASSURANCE	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E- SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand the basics of requirements engineering.					
2.	To learn different techniques used for requirements elicitation.					
3.	To know the role played by requirements analysis in requirement integration.					
4.	To comprehend the use of various methodologies for requirements development.					
5.	To examine the current trends in requirements prioritization and validation.					
UNIT-I	INTRODUCTION	9				
	Introduction to Software Quality - Challenges - Quality Factors - Testing Activities - Test Case Selection - Power of Test - Components of SQA - SQA Components in Project Life Cycle - Test Groups - Software Quality Assurance Group - Reviews.					
UNIT-II	TESTING METHODOLOGIES	9				
	Basics of Software Testing - Test Generation from Requirements - Finite State Models - Combinatorial Designs - Test Selection, Minimization and Prioritization for Regression Testing - Test Adequacy, Assessment and Enhancement.					
UNIT-III	TEST STRATEGIES	9				
	Testing Strategies - White Box and Black Box Approach - Integration Testing - System and Acceptance Testing - Performance Testing - Regression Testing - Internationalization Testing - Ad-hoc Testing - Website Testing - Usability Testing - Accessibility Testing.					
UNIT-IV	TEST AUTOMATION AND MANAGEMENT	9				
	Test plan - Management - Execution and Reporting - Software Test Automation - Automated Testing tools - Hierarchical Models of Software Quality - Configuration Management - Documentation Control.					
UNIT-V	SOFTWARE QUALITY ASSURANCE	9				
	Project progress control - costs - quality management standards - project process standards - management and its role in SQA - SQA unit.					
		Total (L)				45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS					
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments					

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	and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
C01	Develop Quality plans and use SQA components in project life cycle.	L3 – Apply
C02	Demonstrate the process of evaluating product quality.	L3 – Apply
C03	Implement the appropriate infrastructure components and use configuration items for Quality control.	L3 – Apply
C04	Apply various testing methods and verify.	L3 – Apply
C05	Assess Quality standards of various software products.	L3 – Apply
	REFERENCE BOOKS:	
1.	Yogesh Singh, "Software Testing", Cambridge University Press, 2012.	
2.	Daniel Galin, "Software Quality Assurance – from Theory to Implementation" Pearson Education, 2009.	
3.	Aditya Mathur, "Foundations of Software Testing", Pearson Education, 2008.	
4.	Srinivasan Desikan, Gopalaswamy Ramesh, "Software Testing – Principles and Practices", Pearson Education, 2006.	
5.	Ron Patton, "Software Testing", Second Edition, Pearson Education, 2007.	

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
C01	3		3	2		
C02	2		3			
C03	2		2	2		
C04	3		3	3		
C05	2		2	2		3
Average	2.4		2.6	2.2		3.0
1-Low, 2 –Medium, 3-High						


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UX23MC701	UNIVERSAL HUMAN VALUES AND ETHICS	CP	L	T	P	C
		3	1	0	2	NC
Programme & Branch	COMMON TO M.E. (SE,AE,PED,ISE), M.C.A & MBA (IEV)	Version: 1.1				
Course Objectives:						
1.	To understand the concept of Universal Human Values.					
2.	To discuss theoretical and practical implications of UHV.					
3.	To relate the use of harmony in the family and society.					
4.	To classify the harmony in the nature methods.					
5.	To construct effective human values in personal and professional in life.					
UNIT-I	INTRODUCTION TO VALUE EDUCATION					3+6
	Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education) - Understanding Value Education - Sharing about Oneself - Self-exploration as the Process for Value Education - Continuous Happiness and Prosperity - the Basic Human Aspirations - Exploring Human Consciousness - Happiness and Prosperity - Current Scenario - Method to Fulfil the Basic Human Aspirations - Exploring Natural Acceptance . * Experiential Learning: Practice sessions to discuss natural acceptance in human being.					
UNIT-II	HARMONY IN THE HUMAN BEING					3+6
	Understanding Human being as the Co-existence of the Self and the Body - Distinguishing between the Needs of the Self and the Body - Exploring the difference of Needs of Self and Body - The Body as an Instrument of the Self - Understanding Harmony in the Self - Exploring Sources of Imagination in the Self- Harmony of the Self with the Body - Programme to ensure self-regulation and Health - Exploring Harmony of Self with the Body . * Experiential Learning: Practice sessions to discuss the awareness of self, help in managing stress and anxiety.					
UNIT-III	HARMONY IN THE FAMILY AND SOCIETY					3+6
	Harmony in the Family - the Basic Unit of Human Interaction - 'Trust' - the Foundational Value in Relationship - Exploring the Feeling of Trust - 'Respect' - as the Right Evaluation - Exploring the Feeling of Respect - Other Feelings , Justice in Human-to-Human Relationship - Understanding Harmony in the Society - Vision for the Universal Human Order - Exploring Systems to fulfil Human Goal. * Experiential Learning: Case Studies on Family and societal issues.					

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16

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UNIT – IV	HARMONY IN THE NATURE/EXISTENCE	3+6
	Understanding Harmony in the Nature – Interconnectedness , self-regulation and Mutual Fulfilment among the Four Orders of Nature - Exploring the Four Orders of Nature - Realizing Existence as Co-existence at All Levels - The Holistic Perception of Harmony in Existence - Exploring Co-existence in Existence. * Experiential Learning: Case Study to discuss human being as cause of imbalance in nature, pollution, depletion of resources and role of technology.	
UNIT – V	IMPLICATIONS OF THE HOLISTIC UNDERSTANDING - A LOOK AT PROFESSIONAL ETHICS	3+6
	Natural Acceptance of Human Values - Definitiveness of (Ethical) Human Conduct - Exploring Ethical Human Conduct - A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order - Competence in Professional Ethics - Exploring Humanistic Models in Education - Holistic Technologies, Production Systems and Management Models -Typical Case Studies - Strategies for Transition towards Value-based Life and Profession - Exploring Steps of Transition towards Universal Human Order. * Experiential Learning: Case studies of holistic technologies and Professional ethics.	
Total (LP) : 45 Periods		
* Internal Evaluation only.		
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the class room teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Interpret the concepts of Universal Human Values.	L2 - Understand
CO2	Summarize both theoretical and practical implications of Universal Human Values.	L2 - Understand
CO3	Build the harmony in family and society.	L3 - Apply
CO4	Practice harmony in all human existence.	L3 - Apply
CO5	Relate human values in both personal and professional life.	L2 - Understand
	REFERENCE BOOKS:	
1.	R.R Gaur, R Sangal, G P Bagaria, A foundation course in Human Values and professional Ethics – Teachers Manual, Excel books, New	

w.e.f. 13/10/2025

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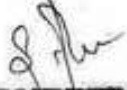
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2.	B L Bajpai, 2004, Indian Ethos and Modern Management, New Royal Book Co., Lucknow, Reprinted 2008.		
3.	Frankl, Viktor E. Yes to Life In spite of Everything, Penguin Random House, London, 2019.		
4.	Van Zomeren, M., & Dovidio, J. F. The Oxford Handbook of the Human Essence (Eds.), New York Oxford University Press, 2018.		
5.	R R Gaur, R Asthana, G P Bagaria, A Foundation Course in Human Values and Professional Ethics, 2nd Revised Edition, Excel Books, New Delhi, 2019.		
6.	A.N. Tripathi, Human Values, New Age Intl. Publishers, New Delhi, 2004.		
S.No.	WEB REFERENCES:		
1.	https://www.researchgate.net/publication/382495858_Universal_Human_Value		
2.	https://fdp-sl.aicte-india.org/UHVII.php		
	VIDEO REFERENCES:		
S.No.	Publisher	Website link	Type of Content
1.	Dr.Rajneesh Arora	https://www.youtube.com/watch?v=SdXIB0k7klQ&list=PLWDeKF97v9SP7wSlapZcQRrT7OH0ZIGC4&index=2	Video Lecture
2.	Mr.Umesh Jadhav	https://www.youtube.com/watch?v=_g8Hw2APx8Q	Video Lecture


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UX23PT802	RESEARCH PAPER REVIEW AND PRESENTATION	CP	L	T	P	C
		2	0	0	2	1
Programme & Branch	Common to all M.E (ISE, AE, PED, SE) and M.C.A.	Version : 1.0				
Course Objectives:						
1.	To learn scientific paper reading and wiring skills.					
2.	To learn the literature review and report wiring skills.					
3.	To understand the research gap and formulation of the research problem.					
The work involves the following steps:						
1 Assigning the faculty supervisor						
2 Selecting a subject, narrowing the subject into a topic						
3 Stating an objective.						
4 Collecting the relevant bibliography (atleast 20 research papers)						
5 Studying the papers understanding the authors contributions and critically analyzing each paper.						
6 Preparing a 20-25 page literature review report						
7 Preparing conclusions based on the literature review report.						
8 Writing the Final Review Paper						
9 Final Presentation to the review committee						
Evaluation method:						
A faculty supervisor will be assigned to each student. The supervisor will assign a topic to the student. The student has to review the literature pertaining to the topic, prepare a 20-25 page report and make a presentation. Minimum 20 research papers have to be reviewed out of which 60% have to be in the recent 05 years. The format for the research paper report and guidelines for assessment will be issued by the Head of the Department before the commencement of the course. The evaluation will be carried out based on the research paper report and presentation, and is evaluated for 100 marks. Minimum 50 marks is essential to pass. In case a student fails, he or she has to redo the course in the forthcoming semesters. Assessment is by Internal Assessment mode only no End Semester Examination.						
Total : 30 Periods						
COURSE OUTCOMES:						BLOOM'S Taxonomy
Upon completion of this course the students will be able to:						
CO1	Write a scientific review paper in their field					L3 - Apply
CO2	Identify the research gap and formulate the research problem					L3 - Apply

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ME23SE309	SOFTWARE AGENTS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E- SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand agent-based systems for performing specific tasks in real-world environments.					
2.	To provide knowledge for designing applications that incorporate key security mechanisms for intelligent agents.					
3.	To apply agent-based technologies in building and managing distributed information systems.					
4.	To familiarize interfacing software agents using standard Agent Communication Languages.					
5.	To apply appropriate technologies and strategies for resolving security challenges in agent-based systems.					
UNIT-I	AGENTS – OVERVIEW	9				
	Agent Definition – Agent Programming Paradigms – Agent Vs Object – Aglet – Mobile Agents – Agent Frameworks – Agent Reasoning.					
UNIT-II	JAVA AGENTS	9				
	Processes – Threads – Daemons – Components – Java Beans – ActiveX – Sockets – RPCs – Distributed Computing – Aglets Programming – Jini Architecture – Actors and Agents – Typed and proactive messages.					
UNIT-III	MULTIAGENT SYSTEMS	9				
	Interaction between agents – Reactive Agents – Cognitive Agents – Interaction protocols – Agent Coordination – Agent negotiation – Agent Cooperation – Agent Organization – Self-Interested agents in Electronic Commerce Applications.					
UNIT-IV	INTELLIGENT SOFTWARE AGENTS	9				
	Interface Agents – Agent Communication Languages – Agent Knowledge Representation – Agent Adaptability – Belief Desire Intension – Mobile Agent Applications – Introduction to Chat Bot					
UNIT-V	APPLICATION OF AGENTS	9				
	Workflow and Business Process Management – Information Retrieval and Management – Electronic Commerce – Human-Computer Interface – Virtual Environments					
		Total (L)				45 Periods

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Develop an agent based system for a particular task.	L3 – Apply
CO2	Design an application that uses different security issues for intelligent agents.	L3 – Apply
CO3	Apply agent-based technologies in the development and application of distributed information systems that use software agents.	L3 – Apply
CO4	Interface agents through Agent communication languages.	L3 – Apply
CO5	Apply technologies for solving agent security issues.	L3 – Apply
	REFERENCE BOOKS:	
1.	Bradshaw, "Software Agents ", MIT Press, 2010.	
2.	Michael Wooldridge, "An Introduction to Multiagent Systems", Second Edition, Wiley, 2009.	
3.	Richard Murch, Tony Johnson, "Intelligent Software Agents", Prentice Hall, 2000.	
4.	Gerhard Weiss, "Multi Agent Systems – A Modern Approach to Distributed Artificial Intelligence", MIT Press, 2000.	
5.	Russel, Norvig, "Artificial Intelligence: A Modern Approach", Second Edition, Pearson Education, 2003.	
6.	Bigus & Bigus, "Constructing Intelligent agents with Java ", Wiley, 1997.	

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3		3	2	3	
CO2			3	3	3	3
CO3				2	3	3
CO4	3			3		3
CO5	2			2	3	
Average	2.6		3.0	2.4	3.0	3.0
1-Low, 2 –Medium, 3-High						

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ME23SE601	PROJECT WORK – PHASE I	CP	L	T	P	C
		12	0	0	12	6
Programme & Branch	M.E. – SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To help students identify real-world engineering problems through literature survey and analysis.					
2.	To develop the ability to plan and execute an engineering project independently.					
3.	To improve skills in technical report writing and project documentation.					
4.	To encourage the application of engineering concepts to practical design or case study problems.					
5.	To enhance communication and presentation skills through reviews and viva-voce examination.					
COURSE CONTENT						
	Each student will individually work on a project topic related to engineering design or manufacturing applications. The topic may be theoretical or based on a case study and must be selected after a detailed literature survey. The chosen topic should be approved by the Head of the Department and carried out under the guidance of a faculty member with expertise in the selected area. The student must submit a project proposal for approval and clearly define the problem, objectives, and methodology. During the course, three project review meetings will be conducted by the Project Review Committee to evaluate progress and provide guidance and suggestions for improvement. At the end of the semester, the student must submit a detailed Phase-I report, including the problem definition, literature review, and proposed methodology. The final evaluation will include continuous assessment during reviews and a viva-voce examination conducted by a panel of examiners, including one external examiner.					
	Total(P)					180 Periods
	Course Outcomes: Upon completion of this course, the students will be able to:					BLOOM'S Taxonomy
CO1	Identify and define an engineering problem based on a detailed literature survey.					L2 - Understand
CO2	Analyze existing research and technical information related to the selected project topic.					L4 - Analyze
CO3	Design an appropriate methodology to solve the identified engineering problem.					L3 - Apply
CO4	Develop a technical project report with proper structure and documentation.					L6 - Create
CO5	Present and defend the project work effectively during reviews and viva-voce examination.					L3 - Apply

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Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	3	2	3	3	3	2
CO2	3	2	3	3	3	2
CO3	3	2	3	3	3	2
CO4	3	3	3	2	3	2
CO5	3	3	3	2	3	2
Avg.	3.0	2.4	3.0	2.6	3.0	2.0
1-Low,2-Medium,3-High						




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ME23SE602	PROJECT WORK – PHASE II	CP	L	T	P	C
		24	0	0	24	12
Programme & Branch	M.E. – SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To implement and complete the approved project work by applying concepts learned.					
2.	To develop the ability to implement, test, and validate engineering solutions.					
3.	To enhance research skills through technical writing and paper publication.					
4.	To improve problem-solving and decision-making skills through continuous project reviews.					
5.	To strengthen technical communication skills through report writing and viva-voce examination.					
COURSE CONTENT						
	This phase is a continuation of the Phase I project and will be carried out on the same topic under the guidance of the same supervisor. The student will continue the work based on the approved methodology and complete the project to the satisfaction of the supervisor and the Project Review Committee. Three project review meetings will be conducted during the semester to monitor progress and provide suggestions for improvement. The student is expected to publish at least one research paper related to the project in a national or international conference. At the end of the course, the student must submit a detailed project report to the Head of the Department. The final evaluation will be based on the project report and a viva-voce examination conducted by a panel of examiners, including one external examiner.					
	Total(P)					360 Periods
	Course Outcomes: Upon completion of this course, the students will be able to:					BLOOM'S Taxonomy
CO1	Implement the approved project plan and carry out the project work successfully.					L3 - Apply
CO2	Analyze project results and evaluate system performance or outcomes.					L4 - Analyze
CO3	Design and implement improvements based on review committee feedback.					L5 - Evaluate
CO4	Prepare a complete technical report and research paper suitable for publication					L6 - Create
CO5	Present and defend the project work effectively during reviews and viva-voce.					L3 - Apply

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Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
C01	3	2	3	3	3	2
C02	3	2	3	3	3	2
C03	3	2	3	3	3	2
C04	3	3	3	2	3	2
C05	3	3	3	2	3	2
Avg.	3.0	2.4	3.0	2.6	3.0	2.0
1-Low,2-Medium,3-High						



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ME23SE401	FRONTEND DEVELOPMENT	CP	L	T	P	C
		5	3	0	2	4
Programme & Branch	M.E. SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To develop interactive web solutions by applying ES6 features in JavaScript.					
2.	To apply UI/UX design principles and techniques to build user- friendly interfaces.					
3.	To gain knowledge in building web applications using React.					
4.	To apply state management concepts by implementing Redux and Redux Toolkit in web applications.					
5.	To build websites using Next.js for search engines optimization.					
UNIT-I	ADVANCED JAVASCRIPT	9				
	Overview of Core JavaScript Features – ES6+: Let and const scope – Arrow functions – Template literals – Destructuring – Spread and Rest operators – Default parameters. Functions and Callbacks: Higher-order functions – Callback functions – Closures-Anonymous and named functions – Function scope vs block scope. Advanced Array Methods: map – filter – reduce. Asynchronous JavaScript: setTimeout() – setInterval() – Event loop and call stack basics – Callbacks vs Promises – Creating and Chaining Promises – async/await, Error handling in Promises. Working with APIs: fetch() API – Handling JSON responses. Introduction to JavaScript Frameworks and Libraries.					
UNIT-II	UI/UX DESIGN	9				
	Overview of Design Thinking – Introduction to UI/UX: Importance of User Interface and User Experience – UI vs UX: Key Differences – Fundamental Principles of Good Design – Figma Tool Overview: Introduction to Figma and Interface Walkthrough – Creating Mindmaps – Designing Wireframes: Low-Fidelity – Mid-Fidelity – High-Fidelity – Building Mockups and Interactive Prototypes.					
UNIT-III	REACT JS	9				
	Introduction to React – Virtual DOM – Setting up the React Development Environment – Components: Functional Class – Props – State Management – Conditional Rendering – Event Handling in JSX – Controlled vs Uncontrolled Components – Form Submission – React Hooks: use State Hook – use Effect Hook – use Context Hook – Lifecycle simulation with Hooks – Rules of Hooks – Fetch API – Axios – Handling loading and error states – React Router Basics – Route – Link – Navigation – Dynamic Routing.					

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UNIT-IV	REDUX	9
	Introduction: Redux – Need for State Management – Store – Actions – Reducers – Setting up Redux in React App – Redux Flow – Provider Component – useSelector Hook – useDispatch Hook – Connecting Redux to Components – Middleware & Async Handling: Redux Middleware – Redux Thunk – Async Actions – Handling API Calls with Redux.	
UNIT-V	INTRODUCTION TO NEXT.JS AND PROJECT DEVELOPMENT	9
	Next.js: Differences from React – File-based Routing – Pages and Navigation – Static and Server-side Rendering – API Routes – Data Fetching (getStaticProps, getServerSideProps) – Styling in Next.js – SEO with Head component – Image Optimization.	
	Total (L)	45 Periods
	LIST OF EXPERIMENTS:	
1.	Create a student result tracker that accepts student marks, calculates average, determines pass/fail.	
2.	Build a Movie Search Web App where users can search films by title and view posters, ratings, and descriptions.	
3.	Create Mid or High – Fidelity Wireframes in Figma (Mobile and Desktop) and Develop an Interactive Clickable Prototype.	
4.	Develop a React application that manages Task & Productivity Tracker with API integration, routing and forms.	
5.	Create a blog homepage and individual post pages with dynamic Routing.	
6.	Build a React App with Global State Management using Redux.	
7.	Build a Next.js App with File-based Routing and Dynamic Pages.	
	Total (P)	30 Periods
	Total (L+P)	75 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Develop interactive web solutions by applying ES6 features in JavaScript.	L3 - Apply
CO2	Apply UI/UX design principles and techniques to create user-friendly interfaces.	L3 - Apply

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C03	Demonstrate React concepts in creating web applications.	L3 - Apply
C04	Apply state management concepts through implementation of Redux and Redux Toolkit in web applications.	L3 - Apply
C05	Build SEO-friendly websites using Next.js to enhance search engine visibility.	L3 - Apply
REFERENCE BOOKS:		
1.	Angel Sola Orbaiceta, "Build a Frontend Web Framework (From Scratch)", 1 st Edition, Manning Publications, 2024.	
2.	Kirupa Chinnathambi, "Learning React: A Hands-On Guide to Building Web Applications Using React and Redux", 2 nd Edition, Addison-Wesley Professional, 2018.	
3.	Joel Marsh, "UX for Beginners: A Crash Course in 100 Short Lessons", 1 st Edition, O'Reilly, 2022.	
4.	Alex Banks, Eve Porcello, "Learning React: Functional Web Development with React and Redux", 1 st Edition, O'Reilly, 2020.	
5.	Shama Hoque, "Full-Stack React Projects", 2 nd Edition, Packo Publishing Ltd., 2020.	

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
C01			3	2	1	1
C02		2	2	3	1	1
C03			3	3	2	2
C04			3	3	2	2
C05	2		3	2	1	2
Average	2.0	2.0	2.8	2.6	1.4	1.6
1-Low, 2 -Medium, 3-High						

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ME23SE402	BACKEND DEVELOPMENT WITH JAVA	CP	L	T	P	C
		5	3	0	2	4
Programme & Branch	M.E. SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand the basics of Java backend development using Servlets and JSP.					
2.	To build applications using Spring MVC and template engines.					
3.	To implement JPA and Hibernate for database operations.					
4.	To create and test REST APIs using Spring Boot.					
5.	To deploy Java backend applications and integrate them with databases & cloud platforms.					
UNIT-I	SERVLET AND JSP					9
	Introduction to Web Applications and Java EE - HTTP Protocol and Request/Response Model - Servlet Lifecycle: init(), service(), destroy() - Handling Form Data (GET, POST methods) - Introduction to JSP (Java Server Pages) - JSP Directives, Scripting Elements, Action Tags - MVC Architecture using Servlet and JSP - Session Management: Cookies and HttpSession.					
UNIT-II	JPA WITH HIBERNATE					9
	Introduction to ORM (Object Relational Mapping) - Hibernate Configuration and Architecture - Entity Classes and Hibernate Annotations - Performing CRUD Operations with Hibernate - Relationships: One-to-One, One-to-Many - Introduction to JPA and JPQL.					
UNIT-III	SPRING CORE AND SPRING MVC					9
	Introduction to Spring Framework - Dependency Injection and Inversion of Control (IoC) - Bean Configuration and Lifecycle - Spring MVC Architecture - Request Mappings and Controllers - Introduction to Thymeleaf Template Engine - Setting up Thymeleaf with Spring MVC - Data Binding and Dynamic HTML Rendering using Thymeleaf - Form Handling and Validation using Spring MVC.					
UNIT-IV	SPRING BOOT					9
	Overview of Spring Boot vs Traditional Spring - Spring Initializer Setup - Auto Configuration and Starter Dependencies - Spring Boot DevTools for Fast Development - Profiles and Property Management - Packaging and Running Spring Boot Applications - Spring Boot Database Connection (JPA Integration).					
UNIT-V	REST API DEVELOPMENT USING SPRING BOOT					9
	REST Architecture and Principles - Building REST APIs (GET, POST, PUT, DELETE) - Request Handling and Response Entities -					

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	Exception Handling in REST Services –Introduction to API Authentication (JWT Overview) – Testing APIs using Postman.	
	Total (L + P)	45 Periods
	LIST OF EXPERIMENTS:	
1.	Build Servlet applications, create dynamic web pages using JSP, implement a Login system using Servlet and JSP	
2.	Build a Java application using Hibernate ORM, Integrate JPA with MySQL Database, Perform CRUD and Relationship operations	
3.	Build a Spring MVC Application, Handle form data with Thymeleaf Templates, Dynamic rendering using Spring and Thymeleaf	
4.	Build a CRUD Application with Spring Boot, Connect and manage databases in Spring Boot Apps	
5.	Develop a complete RESTful API, Test APIs using Postman	
	Total (P)	30 Periods
	Total (L + P)	75 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Describe the process of creating Java web applications using Servlets and JSP.	L2 – Understand
CO2	Explain how data is managed using JPA and Hibernate.	L2 – Understand
CO3	Design modular applications using Spring Core and MVC architecture.	L3 – Apply
CO4	Create and deploy RESTful APIs using Spring Boot.	L3 – Apply
CO5	Implement security measures and perform testing of Java backend applications using appropriate tools and techniques.	L3 – Apply
	REFERENCE BOOKS:	
1.	K. Somasundaram, P. Suresh Kumar, "Java Programming and Web Development", Tata McGraw-Hill, 2019.	
2.	Herbert Schildt, "Java: The Complete Reference", 12 th Edition, McGraw-Hill, 2022.	
3.	Ramesh Fadatore, "Spring 5 Design Patterns", Packt Publishing, 2018.	
4.	Craig Walls, "Spring in Action", 6 th Edition, Manning Publications, 2022.	
5.	Christian Bauer, Gavin King, Gary Gregory, "Java Persistence with Hibernate", 2 nd Edition, Manning Publications, 2015.	

w.e.f. 22/12/2025

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6.	Rod Johnson, Juergen Hoeller, Alef Arendsen, Thomas Risberg, Colin Sampaleanu, "Professional Java Development with the Spring Framework", Wrox, 2009.
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Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1			3	2	1	1
CO2			3	3	2	1
CO3			3	3	2	2
CO4			3	3	3	2
CO5	2		2	2	3	2
Average	2.0		2.8	2.6	2.2	1.6
1-Low, 2 -Medium, 3-High						



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ME23SE403	FULL STACK DEVELOPMENT AND DEPLOYMENT	CP	L	T	P	C
		5	3	0	2	4
Programme & Branch	M.E. SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand and use REST APIs, HTTP requests, CORS, and environment settings for frontend-backend communication.					
2.	To apply secure login and authorization using JWT, user roles, protected pages, and token handling.					
3.	To analyze and develop full-stack applications by connecting frontend forms with backend CRUD APIs using a clean structure and proper error handling.					
4.	To deploy frontend and backend applications with secure settings, correct API links, and production best practices.					
5.	To understand modern development practices such as containerization and basic CI/CD workflows for complete web solutions.					
UNIT-I	FRONTEND-BACKEND COMMUNICATION BASICS					9
	Introduction to REST API Integration - HTTP Requests from Frontend (fetch, axios) - Understanding CORS (Cross-Origin Resource Sharing) and Solutions - API Authentication Tokens (Introduction to JWT) - Environment Variable Management (Frontend side).					
UNIT-II	USER AUTHENTICATION AND AUTHORIZATION					9
	Authentication Flows (Sign up, Login, Token Handling) - Authorization Flows (Role-based access control basics) - Building a Secure Login System (Frontend + Backend) - Handling Protected Routes on Frontend - Token Expiry and Auto-Logout Handling.					
UNIT-III	FULL STACK APPLICATION DEVELOPMENT					9
	Full Stack Project Planning (ERD + Frontend and Backend Sync) - Folder Structure for Full Stack Applications - Connecting Frontend Forms to Backend CRUD APIs - Error Handling and User Feedback (Frontend) - Optimizing API Calls and Code Refactoring.					
UNIT-IV	DEPLOYMENT OF FRONTEND AND BACKEND					9
	Deployment of Frontend Apps - Deployment of Spring Boot Backend Apps - Managing Environment Variables securely - Handling Domain Linking and API URLs - Best Practices for Production Deployment.					
UNIT-V	INTRODUCTION TO DEVOPS BASICS					9
	Introduction to Docker Concepts - Building Simple Docker Containers (Spring Boot Backend) - Awareness of CI/CD Pipelines (GitHub Actions overview) - Version Controlling Deployment.					
	Total (L)					45 Periods

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	LIST OF EXPERIMENTS:	
1.	Create a frontend app fetching data from backend API.	
2.	Handle API token storage (local Storage/session Storage)	
3.	Build a login + protected dashboard full stack mini app.	
4.	Build a small full stack CRUD application.	
5.	Integrate API error handling and toast notifications.	
6.	Deploy full stack app live.	
7.	Solve CORS, domain issues, secure API keys.	
8.	Create a basic Dockerfile for Spring Boot app.	
9.	Setup auto-deploy from GitHub to Vercel for frontend.	
	Total (P)	30 Periods
	Total (L+P)	75 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Integrate React/Angular/Next.js frontend apps with Spring Boot REST APIs.	L2 – Understand
CO2	Apply user authentication and authorization in full stack applications.	L3 – Apply
CO3	Deploy full stack applications with separated frontend and backend hosting.	L3 – Apply
CO4	Execute techniques to resolve deployment issues involving CORS, environment variables, and API proxying.	L3 – Apply
CO5	Interpret the basic concepts of Docker, GitHub Actions and basic CI/CD practices.	L2 – Understand
	REFERENCE BOOKS:	
1.	David Clinton, Tracy Wallace, "Practical GitHub Actions", O'Reilly Media 2023.	
2.	Robin Wieruch, "The Road to React", 2020.	
3.	K. Somasundaram, P. Suresh Kumar, "Java Programming and Web Development", Tata McGraw-Hill, 2019.	
4.	Craig Walls, "Spring in Action", 5 th Edition, Manning Publications, 2018.	
5.	Ramesh Fadatare, "Spring 5 Design Patterns", Packt Publishing, 2018.	

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Passed in BoS Meeting held on 09/12/2025

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Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1			3	2	2	2
CO2			2	3	2	2
CO3			2	2	2	3
CO4			2	3	2	2
CO5	2		2	2	2	3
Average	2.0		2.2	2.4	2.0	2.4
1-Low, 2 -Medium, 3-High						



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ME23SE404	FUNDAMENTALS OF WEB DEVELOPMENT	CP	L	T	P	C
		5	3	0	2	4
Programme & Branch	M.E. SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand the fundamentals of Computer Networks and the Internet.					
2.	To design web pages with media and graphics.					
3.	To develop visual presentation and responsive web pages using CSS.					
4.	To create an interactive web page using advanced JavaScript.					
5.	To develop responsive web layouts using bootstrap and work with jQuery for dynamic web interactions.					
UNIT-I	INTRODUCTION TO WEB DESIGN					9
	Computer Network Basics - Introduction to Internet - History- WWW - Evolution of Web - Web Browser - URI - Web Server - Website - Web System Architecture Single Tier, Two Tier, Three Tier Architecture - HTTP Request and Response - Port - DNS - Introduction to Client-side and Server-side scripting.					
UNIT-II	HTML					9
	Basic HTML Elements - Attributes - Hyperlinks and bookmarks - Images - Favicon - Emojis - List and Table - Form Elements - Layouts - Other HTML5 Elements - Inline Vs Block level elements - Media and Graphics.					
UNIT-III	CSS					9
	Basics of CSS - Types - Selectors - Pseudo classes - Core CSS3 Properties - CSS3 Features - Media Queries CSS Preprocessors (SASS, SCSS).					
UNIT-IV	JAVASCRIPT BASICS					9
	Introduction to JavaScript - HTML Vs CSS Vs JavaScript - Syntax - First JavaScript - Keywords - Variables - Operators - Data Types - Type Conversion - Control Flow Statements - Functions - Event listener - Constructor - Prototypes - Objects: String, Array, Math, Date, Regular Expressions, Events - DOM - Form Validation - Canvas.					
UNIT-V	BOOTSTRAP AND JQUERY					9
	Basics of Bootstrap - Grid System and Layout - Navbar and Navigation Components - Buttons and Alerts - Forms and Input Groups - Cards and Modals - Create Responsive Web Page - Introduction to jQuery - Selecting and Manipulating DOM Elements - Event Handling - jQuery Plugins.					
	Total (L)					45
						Periods

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	LIST OF EXPERIMENTS:	
1.	Acquaintance with elements, Tags and basic structure of HTML files.	
2.	Practicing basic and advanced text formatting.	
3.	Designing of webpage - Document Layout, Forms, Layouts, Lists, Frames, Tables, Controls.	
4.	Acquaintance with creating style sheet, CSS properties and styling.	
5.	Working with HTML elements box, Positioning and Block properties in CSS	
6.	Designing with cascading style sheet-Internal & External Style Sheet	
7.	JavaScript: Designing a Simple Project using Java Script	
	Total (P)	30 Periods
	Total (L+P)	75 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Understand the concepts of Computer Networks and the Internet.	L2 - Understand
CO2	Develop a web page using HTML elements, forms, tables and multimedia components.	L3 - Apply
CO3	Design responsive, visually appealing, and maintainable web pages using CSS.	L3 - Apply
CO4	Apply Javascript objects to build dynamic web page with validation.	L3 - Apply
CO5	Develop a responsive web page using Bootstrap's and integrate jQuery plugins to enhance website features.	L3 - Apply
	REFERENCE BOOKS:	
1.	Philip Ackermann, Full Stack Web Development: The Comprehensive Guide, Rheinwek Computing, First Edition, 2023.	
2.	John Dean, "WEB PROGRAMMING with HTML5, CSS and JavaScript", First Edition, Jones & Bartlett Learning, 2018.	
3.	Nabendu Biwas, Ultimate Full-Stack Web Development with MERN, Orange Education Pvt Ltd., 2023.	
4.	Jennifer Robbins, "Learning Web Design: A Beginner's Guide To HTML, CSS, JavaScript, And Web Graphics", Fifth Edition, O'Reilly Publisher, 2018.	

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5.	A. Powell, "HTML & CSS: The Complete Reference Thomas" Fifth Edition, Tata McGraw Hill, 2010.
6.	Lea Verou, "CSS Secrets: Better Solutions to Everyday Web Design Problems", First Edition, O'Reilly Publisher, 2015.

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3	3		2
CO2	2		3	3		2
CO3	2		3	3		2
CO4	2		3	3		2
CO5			3	2	3	
Average	2.0		3.0	2.8	3.0	2.0
1-Low, 2 -Medium, 3-High						



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ME23SE405	TESTING FUNDAMENTALS AND MANUAL TESTING	CP	L	T	P	C
		5	3	0	2	4
Programme & Branch	M.E. SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand the fundamentals, need and process of software testing.					
2.	To learn various SDLC models and software testing methodologies.					
3.	To apply test case design techniques and prepare necessary documentation.					
4.	To understand bug life cycle and explore test management tools and techniques.					
5.	To apply agile testing principles and manage test projects using tools like Jira.					
UNIT-I	INTRODUCTION TO SOFTWARE TESTING					9
	Software Testing Introduction – Importance of Software Testing – Errors, Bugs, Defects, Failures – Principles of Software Testing – SDLC Phases – Role of Testing in SDLC – Overview of Verification and Validation – Introduction to STLC.					
UNIT-II	SDLC MODELS AND TESTING METHODOLOGIES					9
	Waterfall Model – V-Model – Agile Model – Iterative Model – Spiral Model – Big Bang Model – Introduction to Software Testing Methodologies – White Box Testing – Black Box Testing – Grey Box Testing – Static vs Dynamic Testing – Comparison of SDLC Models.					
UNIT-III	TEST CASE DESIGN TECHNIQUES AND DOCUMENTATION					9
	Test Case vs Test Scenario – Test Case Design Techniques: Boundary Value Analysis – Equivalence Partitioning – Decision Table – State Transition – Use Case Testing – Requirement Traceability Matrix (RTM) – Test Plan Preparation – Entry and Exit Criteria – Test Case Template.					
UNIT-IV	BUG LIFE CYCLE AND TEST MANAGEMENT					9
	Defect/Bug Life Cycle – Severity and Priority – Bug Report Preparation – Logging Bugs with Screenshots – Different Test Suites and Types: Smoke, Sanity, Regression, Retesting, Alpha and Beta Testing – Overview of QA and QC – Roles in Testing Team – Communication with Developers.					
UNIT-V	AGILE TESTING AND TESTING PROJECT					9
	Agile Methodology Overview – Scrum Roles and Ceremonies – Writing User Stories and Acceptance Criteria – Introduction to Jira for Test Management – Logging Test Cases and Bugs – Sprint Board and Backlog Management. Testing Project: Writing Effective Test Cases – Defect Reports – Traceability Matrix – Hands-on with Test Management in Jira.					
	Total (L)					45 Periods

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	LIST OF EXPERIMENTS:	
1.	Identifying Errors, Bugs, Defects, and Failures in a Sample Code.	
2.	Comparison of SDLC Models with Focus on Testing Phase.	
3.	Design Test Cases using Boundary Value Analysis and Equivalence Partitioning.	
4.	Bug Report Creation and Classification Based on Severity and Priority.	
5.	Managing Test Cases and Bugs Using Jira.	
	Total (P)	30 Periods
	Total (L+P)	75 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Explain software testing concepts, principles, and its role in SDLC.	L2 - Understand
CO2	Differentiate between testing methods and SDLC models.	L2 - Understand
CO3	Apply different design test cases using standard techniques and document testing artifacts.	L3 - Apply
CO4	Manage defects using bug life cycle concepts and differentiate various testing types.	L2 - Understand
CO5	Implement Agile testing concepts and use Jira for test case management in real-time projects.	L3 - Apply
	REFERENCE BOOKS:	
1.	Srinivasan Desikan and Gopalaswamy Ramesh, "Software Testing: Principles and Practices", 2 nd Edition, Pearson Education, 2005.	
2.	Dorothy Graham, Rex Black, Erik van Veenendaal, "Foundations of Software Testing: ISTQB Certification", 4 th Edition, Cengage Learning, 2020.	
3.	Ron Patton, "Software Testing", 2 nd Edition, Sams Publishing, 2005.	
4.	William E. Perry, "Effective Methods for Software Testing", 3 rd Edition, Wiley, 2006.	
5.	Paul C. Jorgensen, "Software Testing: A Craftsman's Approach", 4 th Edition, CRC Press, 2013.	

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Passed in BoS Meeting held on 09/12/2025

Approved in Academic Council Meeting held on 15/12/2025

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3	2	3	1
CO2	2		3	3	3	1
CO3		2	3	3	3	2
CO4		2	3	2	3	2
CO5	1	2	3	3	3	3
Average	1.6	2.0	3.0	2.6	3.0	1.8
1-Low, 2 -Medium, 3-High						



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 Passed in BoS Meeting held on 09/12/2025
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ME23SE406	AUTOMATION TESTING WITH SELENIUM	CP	L	T	P	C
		5	3	0	2	4
Programme & Branch	M.E. SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand the fundamentals of automation testing and compare it with manual testing to determine suitable testing scenarios.					
2.	To explore Selenium and its components to identify and interact with various web elements using locators like XPath, CSS Selectors.					
3.	To implement browser-based automation using Selenium WebDriver to handle user actions, alerts, frames, and multiple browser windows.					
4.	To apply data-driven testing techniques using Excel/flat files and manage synchronization issues using appropriate wait strategies.					
5.	To design structured test automation frameworks using Page Object Model (POM), incorporate logging and error handling mechanisms for real-time testing.					
UNIT-I	INTRODUCTION TO AUTOMATION TESTING AND SELENIUM					9
	Introduction to Automation Testing – Manual Testing vs Automation Testing – When to go for Automation – Overview of different Automation Tools – Advantages and Limitations of Automation Testing – Introduction to Selenium – Features of Selenium – Selenium Components and Architecture – Installing Selenium and Configuring Eclipse IDE.					
UNIT-II	WEB ELEMENTS AND LOCATORS IN SELENIUM					9
	Basics of HTML Web Elements – Types of Selenium Locators: id, className, name, linkText, partialLinkText, tagName, CSS Selector, XPath – Writing and using XPath effectively – Introduction to SelectorsHub – Inspecting and locating UI elements – Handling Dynamic Elements.					
UNIT-III	SELENIUM WEBDRIVER AND USER ACTIONS					9
	Launching browsers using WebDriver – Working with web elements: click, sendKeys, getText, isDisplayed, clear – Handling alerts and popups – Handling frames and nested frames – Switching between multiple windows and tabs – Mouse hover and keyboard operations using Actions class – Handling Tooltips and Drag-and-Drop actions.					
UNIT-IV	DATA-DRIVEN TESTING AND SYNCHRONIZATION					9
	What is Data-Driven Testing – Creating data-driven tests using Excel and Flat files – Apache POI Basics – Synchronization in Selenium: Implicit Wait, Explicit Wait, Fluent Wait – Managing page load times – Taking screenshots on failures – Handling cookies and JavaScript popups.					

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Passed in BoS Meeting held on 09/12/2025

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UNIT-V	SELENIUM FRAMEWORK AND REAL-TIME TESTING	9
	Introduction to Selenium Frameworks – Page Object Model (POM) – Page Factory implementation – Data-driven testing using Excel and Flat Files – Screenshot capture on failure – Handling Tooltips and Drag-and-Drop elements – Config Properties – Log4j logging– Handling Cookies.	
	Total (L)	45 Periods
	LIST OF EXPERIMENTS:	
1.	Install and Configure Selenium with Eclipse IDE.	
2.	Automate Web Element Identification Using Various Locators.	
3.	Handle Frames, Alerts, and Perform Mouse/Keyboard Actions.	
4.	Data-Driven Login Test Using Excel and Apache POI.	
5.	Create a Mini Selenium Framework Using Page Object Model (POM).	
	Total (P)	30 Periods
	Total (L+P)	75 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Explain the concepts of automation testing, differentiate it from manual testing, and identify appropriate scenarios for automation.	L2 - Understand
CO2	Identify and apply various Selenium locators and WebDriver commands to interact with web elements.	L3 - Apply
CO3	Demonstrate the ability to handle frames, alerts, windows, and perform advanced user interactions.	L3 - Apply
CO4	Develop data-driven automated test scripts using Excel/flat files and implement synchronization strategies.	L3 - Apply
CO5	Design and implement maintainable test automation frameworks using POM, Page Factory, and integrate logging.	L3 - Apply
	REFERENCE BOOKS:	
1.	Unmesh Gundecha, "Selenium Testing Tools Cookbook", 2 nd Edition, Ingram short title, 2015.	
2.	Unmesh Gundecha and Satya Avasaral, "Selenium WebDriver 3 Practical Guide", 2 nd Edition, Packt Publishing Ltd, 2018.	
3.	Navneesh Garg, "Test Automation using Selenium WebDriver with Java", Atlantic Publishers and Distributors (P) Ltd, 2014.	

w.e.f. 22/12/2025

Passed in BoS Meeting held on 09/12/2025

Approved in Academic Council Meeting held on 15/12/2025


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4.	Varun Menon, "TestNG Beginner's Guide", Packt Publishing Limited, 2013.
5.	Unmesh Gundecha, "Learning Selenium Testing Tools with Python", Packt Publishing Limited, 2014.

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3	2	3	1
CO2			3	3	3	1
CO3			3	3	3	1
CO4			3	3	3	2
CO5			3	3	3	2
Average	2.0		3.0	2.8	3.0	1.4
1-Low, 2 -Medium, 3-High						



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ME23SE407	DEVOPS AND CLOUD BASICS	CP	L	T	P	C
		5	3	0	2	4
Programme & Branch	M.E. SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand the DevOps lifecycle, tools, CI/CD pipeline structure, and the business value of DevOps practices.					
2.	To enable learners to use Git, Jenkins, and automation workflows for efficient software deployment.					
3.	To understand CI/CD concepts use Jenkins to automate build, test, and basic deployment pipeline stages.					
4.	To apply cloud computing concepts and deploy applications on AWS.					
5.	To build and run containerized applications using Docker.					
UNIT-I	DEVOPS FOUNDATIONS & LINUX ESSENTIALS	9				
	DevOps introduction – Evolution and Need – Agile vs DevOps – DevOps Culture and Principles – DevOps Lifecycle – Continuous Integration / Continuous Delivery / Continuous Testing Overview – Linux OS Architecture – Filesystem Structure – Permissions and Processes – Shell Basics.					
UNIT-II	VERSION CONTROL, GITHUB WORKFLOW & BUILD AUTOMATION	9				
	Version Control Concepts – Git Basics (Clone, Commit, Merge, Branching) – GitHub Collaboration – Pull Request Workflow – Branching Models – Build Tools (Maven/Gradle) – Artifact Generation Basics.					
UNIT-III	CI/CD PIPELINE & JENKINS AUTOMATION	9				
	Continuous Integration Concepts – CI/CD Benefits – Jenkins Architecture & Plugins – Pipeline Jobs – Build Automation – Testing Integration – Basic Deployment Pipeline Stages.					
UNIT-IV	CLOUD COMPUTING & AWS FUNDAMENTALS	9				
	Cloud Computing Introduction – IaaS/PaaS/SaaS – Virtual Machines vs Containers – AWS Global Infrastructure – EC2, IAM, S3, VPC Fundamentals – Cloud Networking & Security Basics.					
UNIT-V	CONTAINERIZATION, KUBERNETES & DEVOPS INTEGRATIONS	9				
	Container Concepts – Docker Architecture, Images & Registries – Dockerfile Basics – Kubernetes Architecture – Pods, Nodes, Services – Intro to Monitoring & Logging – DevOps Best Practices.					
	Total (L)	45 Periods				

w.e.f. 22/12/2025

Passed in BoS Meeting held on 09/12/2025

Approved in Academic Council Meeting held on 15/12/2025

	LIST OF EXPERIMENTS:	
1.	Linux command exercises – File/User permission tasks – Shell scripting introduction – Git setup and repository initialization	
2.	a) Project versioning using Git b) Git branching strategy c) Pull request and merge practice d) Build generation using Maven	
3.	a) Install and configure Jenkins b) Create freestyle & pipeline jobs c) Integrate Git repository with Jenkins d) CI workflow execution for sample Java backend	
4.	a) Create AWS free tier account b) Launch & access EC2 server c) Deploy files to S3 bucket d) Configure IAM & security groups	
5.	a) Docker container build & run b) Docker Hub push c) Kubernetes deployment using Minikube d) End-to-end DevOps deployment mini project	
	Total (P)	30 Periods
	Total (L+P)	75 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Understand DevOps lifecycle, tools, CI/CD pipeline structure and its business value.	L2 – Understand
CO2	Apply Git, Jenkins, and automation workflows for software deployment.	L3 – Apply
CO3	Design and execute CI/CD pipelines using Jenkins to automate build, testing, and basic deployment processes.	L3 – Apply
CO4	Build and run containerized applications using Docker.	L3 – Apply
CO5	Apply Kubernetes clusters, pods, nodes and deploy services.	L3 – Apply
	REFERENCE BOOKS:	
1.	Gene Kim, Jez Humble, Patrick Debois, John Willis, "The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations", IT Revolution Press.	

w.e.f. 22/12/2025

Passed in BoS Meeting held on 09/12/2025

Approved in Academic Council Meeting held on 15/12/2025


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2.	Jez Humble, Gene Kim & Patrick Debois, "The DevOps Handbook: How to Create World-Class Agility, Reliability & Security in Technology Organizations", O'Reilly Media, 2022.
3.	Nigel Poulton, "Docker Deep Dive", 4 th Edition, Leanpub Publishing, 2023
4.	Kelsey Hightower, Brendan Burns & Joe Beda, "Kubernetes: Up and Running – Dive into the Future of Infrastructure", 3 rd Edition, O'Reilly Media, 2022.
5.	Thomas Erl, "Cloud Computing: Concepts, Technology & Architecture", Pearson Education, 2017.
6.	Ramesh Fadatare, "Spring 5 Design Patterns", Packt Publishing, 2018.

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1			3	2	2	2
CO2			2	2	2	3
CO3	2		3	3	3	3
CO4	2		3	2	2	2
CO5	2		3	3	2	3
Average	2.0		2.8	2.4	2.2	2.6

1-Low, 2 –Medium, 3-High



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ME23SE408	AGILE BASED PROJECT AUTOMATION WITH DEVOPS	CP	L	T	P	C
		5	3	0	2	4
Programme & Branch	M.E. SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand Agile methodologies and principles for efficient project management.					
2.	To learn DevOps practices, culture, and automation effectively.					
3.	To understand and apply testing, quality assurance, and continuous monitoring strategies in DevOps.					
4.	To apply collaboration and teamwork for successful Agile and DevOps projects.					
5.	To apply Agile and DevOps in real-world scenarios and enterprise projects.					
UNIT-I	AGILE METHODOLOGIES AND PRINCIPLES	9				
	Agile Methodologies Overview – Agile Frameworks and Practices – Agile Project Management Techniques – Agile Estimation and Planning – Agile Metrics and Performance Monitoring.					
UNIT-II	DEVOPS IMPLEMENTATION AND AUTOMATION	9				
	Introduction to DevOps Principles – DevOps Culture and Values – DevOps Toolchain and Automation Practices – Continuous Integration and Deployment (CI/CD) Pipelines – Infrastructure as Code (IaC) and Configuration Management.					
UNIT-III	TESTING, QUALITY ASSURANCE, AND MONITORING	9				
	Agile Testing Strategies and Techniques – Quality Assurance in Agile and DevOps – Continuous Testing and Integration Best Practices – Continuous Monitoring and Logging – Security and Compliance in DevOps.					
UNIT-IV	COLLABORATION AND TEAMWORK IN DEVOPS	9				
	Collaborative DevOps Teams and Roles – Effective Communication and Collaboration Practices – Agile Development and DevOps Integration – Measuring Success in Agile-Based Projects – Continuous Improvement in Agile and DevOps.					
UNIT-V	REAL-WORLD APPLICATION AND CASE STUDIES	9				
	Real-world Agile and DevOps Case Studies – Strategies for Agile and DevOps Adoption and Implementation – Managing Agile and DevOps Projects Successfully – DevOps Tools, Automation, and Continuous Monitoring – Scaling Agile and DevOps for Enterprise-Level Projects.					
	Total (L)					45 Periods

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Passed in BoS Meeting held on 09/12/2025

Approved in Academic Council Meeting held on 15/12/2025

	COURSE PROJECT:	
1.	To prepare a document the steps taken to implement in any project.	
	Total (P)	30 Periods
	Total (L+P)	75 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Understand the Agile and DevOps practices.	L2 - Understand
CO2	Apply DevOps principles using automation tools in real-world scenarios.	L3 - Apply
CO3	Implement testing, quality assurance and monitoring strategies in the real-time applications.	L3 - Apply
CO4	Demonstrate collaboration and communication strategies in Agile-DevOps team.	L3 - Apply
CO5	Apply and analyze Agile-DevOps strategies based on real-world use cases.	L3 - Apply
	REFERENCE BOOKS:	
1.	Gene Kim, Jez Humble, Patrick Debois, and John Willis, "The DevOps Handbook: How to Create World-Class Agility, Reliability, & Security in Technology Organizations", 2 nd Edition, IT Revolution Press, 2021.	
2.	Ken Schwaber, "Agile Project Management with Scrum", 2 nd Edition, Microsoft Press, 2020.	
3.	David Farley, "Modern Software Engineering: Doing What Works to Build Better Software Faster", 1 st Edition, Addison-Wesley Professional, 2021.	
4.	Richard Hundhausen, "Professional Scrum Development with Azure DevOps", 1 st Edition, Microsoft Press, 2021.	
5.	Stefano Demiliani, Amit Malik, Sjoukje Zaal, "Azure DevOps Explained: Get Started with Practices and Tools", 1 st Edition, Packt Publishing, 2020.	

w.e.f. 22/12/2025

Passed in BoS Meeting held on 09/12/2025

Approved in Academic Council Meeting held on 15/12/2025

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1			3			2
CO2			2			3
CO3			2		3	2
CO4		2				2
CO5	2		3		2	2
Average	2.0	2.0	2.5		2.5	2.2
1-Low, 2 -Medium, 3-High						



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ME23SE409	CI AND CD (CONTINUOUS INTEGRATION AND CONTINUOUS DEVELOPMENT)	CP	L	T	P	C
		5	3	0	2	4
Programme & Branch	M.E. SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand the concepts of CI/CD pipeline setup.					
2.	To setup and manage the components of CI/CD pipeline.					
3.	To explore deployment strategies in software development.					
4.	To integrate CI/CD in Agile and DevOps environment.					
5.	To understand the implementation strategies in real-world scenario.					
UNIT-I	INTRODUCTION TO CI/CD	9				
	Introduction to Continuous Integration (CI) - Introduction to Continuous Development (CD) - Benefits of CI/CD in Software Development - CI/CD Principles and Workflow - CI/CD Tools and Ecosystem.					
UNIT-II	CI/CD FUNDAMENTALS	9				
	Setting Up CI/CD Pipelines - Automated Testing in CI/CD - Version Control Integration - Artifact Management in CI/CD - Code Quality and Code Reviews.					
UNIT-III	ADVANCED CI/CD PRACTICES	9				
	Continuous Deployment and Delivery - Containerization for CI/CD - Infrastructure as Code (IaC) - Blue-Green and Canary Deployments - Security in CI/CD Pipelines.					
UNIT-IV	CI/CD IN AGILE AND DEVOPS	9				
	CI/CD in Agile Methodologies - CI/CD Integration with DevOps - CI/CD Best Practices in DevOps - CI/CD for Microservices Architecture - Monitoring and Metrics in CI/CD.					
UNIT-V	CI/CD CASE STUDIES AND IMPLEMENTATION	9				
	Real-world CI/CD Case Studies - Designing Effective CI/CD Pipelines - Continuous Improvement in CI/CD - Implementing CI/CD in Your Organization - CI/CD Challenges and Solutions.					
		Total (L)				45 Periods
	COURSE PROJECT:					
1.	To integrate CI/CD with DevOps.					
		Total (P)				30 Periods
		Total (L+P)				75 Periods

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Passed in BoS Meeting held on 09/12/2025

Approved in Academic Council Meeting held on 15/12/2025

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Understand the principles and workflow of CI/CD and identify appropriate tools.	L2 - Understand
CO2	Apply and configure CI/CD pipelines with automation and version control integration.	L3 - Apply
CO3	Apply CI/CD practices and integrate security in software deployment.	L3 - Apply
CO4	Analyze the role of CI/CD in Agile and DevOps for microservices and performance monitoring.	L3 - Apply
CO5	Evaluate CI/CD implementations for deployments in real-world environments.	L3 - Apply
	REFERENCE BOOKS:	
1.	Jez Humble and David Farley, "Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation", Pearson Education, 2010.	
2.	Nicole Forsgren, Jez Humble, and Gene Kim, "Accelerate: The Science of Lean Software and DevOps: Building and Scaling High Performing Technology Organizations", 1 st edition, IT Revolution Press, 2018.	
3.	Mandi Walls, "Building a DevOps Culture", O'Reilly Media, 2013.	

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3	2	2	1
CO2			3	2	3	2
CO3			3	2	3	2
CO4	2		3	2	3	3
CO5	3	2	3	2	3	3
Average	2.3	2.0	3.0	2.0	2.8	2.2
1-Low, 2 -Medium, 3-High						

w.e.f. 22/12/2025
Passed in BoS Meeting held on 09/12/2025
Approved in Academic Council Meeting held on 15/12/2025

ME23SE410	ADVANCED TESTING WITH TESTNG AND CI/CD	CP	L	T	P	C
		5	3	0	2	4
Programme & Branch	M.E. SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand the importance of testing frameworks, create and run test cases using TestNG in Eclipse IDE.					
2.	To understand XML file structure of TestNG, Integrate with selenium for End-to-End Automation					
3.	To explore Maven lifecycle and apply Maven for building, executing Selenium and TestNG projects					
4.	To practice GIT commands to create Maven-based automation projects for repository management					
5.	To design and configure CI/CD pipelines in Jenkins with Selenium and TestNG integration for executing automated test workflows.					
UNIT-I	INTRODUCTION TO TESTNG FRAMEWORK					9
	Importance of Testing Frameworks in Automation - Overview: TestNG vs Junit - Installing TestNG Plugin in Eclipse - TestNG Annotations and Execution Flow - Creating and Running a Simple TestNG Test - Grouping Test Cases - Prioritizing and Sequencing Tests - Disabling and Enabling Test Cases.					
UNIT-II	TESTNG ADVANCED FEATURES AND ASSERTIONS					9
	Understanding TestNG XML File Structure - Parameterization using TestNG.xml - Soft vs Hard Assertions - Data Providers for Data-Driven Testing - Handling Failed Tests (Retry Mechanisms) - Integrating TestNG with Selenium for End-to-End Automation - Generating TestNG Reports.					
UNIT-III	BUILD MANAGEMENT WITH MAVEN					9
	Introduction to Maven for Automation Projects - Maven Lifecycle and POM (Project Object Model) - Creating a Maven-Based Selenium and TestNG Project - Adding Dependencies via Maven Central Repository- Running TestNG Tests using Maven (mvn test) - Generating Maven Surefire Reports.					
UNIT-IV	VERSION CONTROL WITH GIT AND GITHUB					9
	Introduction to Version Control - Git and GitHub Overview - Installing Git on Windows - Basic Git Commands: init, clone, add, commit, push, pull - Managing Branches and Merges - Ignoring Files using .gitignore - Connecting Maven Project to GitHub - Best Practices for Commit Messages and Repository Management.					
UNIT-V	ADVANCED CI/CD AND REAL-TIME INTEGRATION					9
	Introduction to CI/CD Concepts - Jenkins Overview and Installation					

w.e.f. 22/12/2025

Passed in BoS Meeting held on 09/12/2025

Approved in Academic Council Meeting held on 15/12/2025

	- Jenkins Configuration for Automation - Integrating Jenkins with Selenium WebDriver - Installing and Using Maven Plugin in Jenkins - Creating Freestyle and Maven Jobs - Integrating Jenkins with GitHub - Publishing HTML and TestNG Reports through Jenkins.	
	Total (L)	45 Periods
	LIST OF EXPERIMENTS:	
1.	Create parameterized tests using testng.xml and Data Providers.	
2.	Implement soft and hard assertions in test automation.	
3.	Execute Selenium + TestNG scripts and generate TestNG HTML reports.	
4.	Create and configure a Maven-based Selenium + TestNG project.	
5.	Add dependencies and manage project builds using Maven.	
6.	Generate Surefire Reports for executed test cases.	
7.	Initialize and manage a local Git repository.	
8.	Push a Maven project to GitHub and manage branches.	
9.	Implement collaborative version control using Git CLI.	
10.	Install and configure Jenkins for CI/CD.	
11.	Integrate Jenkins with GitHub and Maven.	
12.	Automate Selenium + TestNG builds and publish TestNG reports through Jenkins.	
	Total (P)	30 Periods
	Total (L+P)	75 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
C01	Understand the testing frameworks in automation and along with the steps to install and create TestNG tests in Eclipse IDE.	L2 - Understand
C02	Integrate TestNG with Selenium to build end-to-end automated test frameworks for web applications.	L2 - Understand
C03	Manage build automation and project dependencies using Maven for seamless test execution.	L3 - Apply
C04	Implement Maven-based automation project using Git and collaborate effectively using GitHub repositories.	L3 - Apply

w.e.f. 22/12/2025

Passed in BoS Meeting held on 09/12/2025

Approved in Academic Council Meeting held on 15/12/2025

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CO5	Demonstrate the configuration and operation of CI/CD pipelines in Jenkins using Selenium and TestNG for automating continuous testing.	L3 – Apply
REFERENCE BOOKS:		
1.	Pranoday Pramod Dingare, "Selenium and TestNG: A Beginner's Guide", Packt Publishing.	
2.	Boni García, "Mastering Software Testing with TestNG and Selenium", Packt Publishing, 2017.	
3.	Pallavi sharma., "Selenium WebDriver with Java – Basics to Advanced & Interview Guide", BPB Publications, 2021.	
4.	John Ferguson Smart, Jenkins, "The Definitive Guide", O'Reilly Media, 2011.	
5.	Stephen Nelson-Smith, "Test-Driven Infrastructure with Jenkins", O'Reilly Media, 2013.	
6.	Tim O'Brien, "Maven: The Complete Reference", Sonatype, 2010.	

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3	2	3	1
CO2			3	3	3	2
CO3			3	3	3	2
CO4			3	3	3	2
CO5			3	3	3	3
Average	2.0		3.0	2.8	3.0	2.0
1-Low, 2 -Medium, 3-High						

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Passed in BoS Meeting held on 09/12/2025
Approved in Academic Council Meeting held on 15/12/2025

ME23SE411	TOOLS FOR AUTOMATED TESTING	CP	L	T	P	C
		5	3	0	2	4
Programme & Branch	M.E. SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand the behavior driven development evolution and lifecycle scenarios.					
2.	To understand the Gherkin language and test scenarios.					
3.	To learn how to set up and use the Cucumber framework and to run BDD test scenarios.					
4.	To implement the various automated test reporting techniques.					
5.	To learn how to use AutoIt tools for compiling and execution.					
UNIT-I	INTRODUCTION TO BEHAVIOR DRIVEN DEVELOPMENT					9
	Behavior Driven Development: Behavior-Driven Development (BDD) – History and Evolution of BDD – Difference between TDD and BDD – Benefits of BDD for Testers, Developers, and Business Stakeholders – BDD Lifecycle and Flow in Agile/Scrum Environments – Following a BDD Approach – Mapping User Stories to BDD Scenarios.					
UNIT-II	GHERKIN LANGUAGE AND SCENARIO DESIGN					9
	Introduction to Gherkin Language: Introduction to Gherkin Language – Role of Gherkin in BDD – Supported Tools and Languages – Syntax and Structure of Gherkin Files – Gherkin Keywords: Feature, Scenario, Given, When, Then, And, But – Writing BDD Tests using DSL (Domain Specific Language) – Types of Gherkin Scenarios.					
UNIT-III	BDD IN AUTOMATION USING CUCUMBER					9
	Introduction to Cucumber: Introduction to the Cucumber Framework – Setting up Cucumber with Java and Maven – Cucumber Project Structure and Key Components – Writing Step Definitions for Gherkin Scenarios – Executing BDD Scenarios with Cucumber – Parameterizing Scenarios with Scenario Outline and Data Tables.					
UNIT-IV	TEST EXECUTION REPORTS IN AUTOMATED TESTING					9
	Automated Testing: Introduction to Test Reporting – Generating ExtentSpark Reports – ExtentReports Library Overview – Adding ExtentReports Dependency in Maven – Setting Up ExtentSpark Reporter – Logging Test Status (Pass, Fail, Skip, Info) – Attaching Screenshots to Reports – Generating Allure Reports – Allure Reporting Framework Overview – Adding Allure TestNG Dependency – Configuring Allure Annotations (@Step, @Attachment) – Generating Allure Results and Viewing Reports.					

w.e.f. 22/12/2025

Passed in BoS Meeting held on 09/12/2025

Approved in Academic Council Meeting held on 15/12/2025

UNIT-V	AUTOMATED WINDOWS DIALOGS USING AUTOIT	9
	Introduction to AutoIt: Introduction to AutoIt- Installing AutoIt and Overview of Script Editor – Script Structure and Syntax (Comments, Variables, Functions) – Identifying Windows Elements – Working with Key AutoIT Commands – Automating File Upload Operations – Compiling AutoIt Scripts to Executable (.exe) Files.	
	Total (L)	45 Periods
	LIST OF EXPERIMENTS:	
1.	Analyze user stories and derive BDD scenarios.	
2.	Write simple feature descriptions for a sample user story using BDD principles.	
3.	Write multiple Gherkin feature files for login, registration, and search functionalities.	
4.	Practice different scenario types using Given-When-Then structure.	
5.	Create and execute a Cucumber project integrated with Maven.	
6.	Implement step definitions and parameterized scenarios.	
7.	Integrate Cucumber with Selenium for end-to-end test automation.	
8.	Integrate ExtentReports with Cucumber project and generate visual reports.	
9.	Capture screenshots on test failure and attach to reports.	
10.	Configure Allure Reporting and generate HTML reports for automated test suites.	
11.	Write AutoIt scripts to automate Windows-based dialogs and pop-ups.	
12.	Integrate AutoIt with Selenium test cases for file upload automation.	
13.	Compile and execute AutoIt scripts from Java programs.	
	Total (P)	30 Periods
	Total (L + P)	75 Periods

	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	

w.e.f. 22/12/2025

Passed in BoS Meeting held on 09/12/2025

Approved in Academic Council Meeting held on 15/12/2025

	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Explain the principles and workflow of behavior-driven development.	L2 – Understand
CO2	Describe how BDD scenarios are written using Gherkin language	L2 – Understand
CO3	Implement automation frameworks using Cucumber with Java and Maven.	L3 – Apply
CO4	Generate detailed execution reports using ExtentReports and Allure.	L3 – Apply
CO5	Automate Windows-based dialogs and file uploads using AutoIt.	L3 – Apply
REFERENCE BOOKS:		
1.	John Ferguson Smart, "BDD in Action: Behavior-Driven Development for the Whole Software Lifecycle", Manning Publications, 2014.	
2.	Richard Lawrence & Paul Rayner, "BDD with Cucumber: Specification by Example for Java Developers", Packt Publishing, 2019.	
3.	Boni García, "Mastering Software Testing with TestNG and Selenium", Packt Publishing, 2017.	
4.	Gojko Adzic, "Specification by Example: How Successful Teams Deliver the Right Software", Addison-Wesley, 2011.	
5.	Alan Richardson, "Automating and Testing a REST API: A Practical Guide for Testers and Automation Engineers", Compendium Developments, 2018.	

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3	2	3	1
CO2			3	3	3	1
CO3			3	3	3	2
CO4		2	3		3	1
CO5	2		3	3	3	1
Average	2.0	2.0	3.0	2.7	3.0	1.2
1-Low, 2 -Medium, 3-High						

w.e.f. 22/12/2025
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ME23SE412	PERFORMANCE AND DATABASE TESTING	CP	L	T	P	C
		5	3	0	2	4
Programme & Branch	M.E. SOFTWARE ENGINEERING	Version: 1.0				
Course Objectives:						
1.	To understand basics of performance testing and measures how an application works under different load conditions.					
2.	To install and configure JMeter Architecture, utilize to create and manage test simulation.					
3.	To understand GUI and Non-GUI execution modes in JMeter and identify performance bottlenecks HTML reports.					
4.	To differentiate between database testing and UI testing, perform CRUD operations on databases.					
5.	To learn SQL-based verification and database test automation using Java and JDBC.					
UNIT-I	INTRODUCTION TO PERFORMANCE TESTING					9
	Performance Testing Overview - Significance and Scope - Performance vs Functional Testing - Types of Performance Testing: Load, Stress, Spike, Endurance, Volume, Scalability - Key Metrics: Response Time, Throughput, Latency, Error Rate, Concurrency, Resource Utilization - Performance Testing Lifecycle - Integration with Agile and DevOps - Challenges in Performance Validation.					
UNIT-II	PERFORMANCE TESTING TOOLS & TEST DESIGN (JMETER)					9
	Introduction to Performance Testing Tools - Apache JMeter Architecture - Installation & Configuration - Components of Test Plan: Thread Groups, Samplers, Controllers, Listeners, Timers, Assertions, Config Elements - Test Scenario Design - Parameterization (CSV Data Set Config) - Correlation Basics & Dynamic Values - Script Debugging.					
UNIT-III	TEST EXECUTION, MONITORING & REPORTING					9
	GUI vs Non-GUI Execution Modes - Distributed Load Testing Concepts - Server Monitoring: CPU, Memory, Disk I/O, Network Usage - JMeter Plugins Usage - Result File Analysis - Identifying Performance Bottlenecks - HTML Dashboard Generation - Performance Optimization Approaches.					
UNIT-IV	INTRODUCTION TO DATABASE TESTING					9
	Database Testing Overview - Database vs UI Testing - Types: Structural, Functional, Integrity & Validation Testing - RDBMS Architecture - SQL Essentials - CRUD Operations - Joins & Subqueries - Views, Indexes, Stored Procedures, Triggers - Constraints - Normalization Concepts.					
UNIT-V	DATABASE TESTING TECHNIQUES & AUTOMATION					9
	Schema Validation - Data Type & Size Validation - Constraint & Relationship Testing - Data Migration/ETL Testing - DB Security					

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	Testing Basics – DB Performance Testing Concepts – Writing SQL for Verification – Automation using Java + JDBC Framework Overview.	
	Total (L)	45 Periods
	LIST OF EXPERIMENTS:	
1.	a) Hands-on with performance concepts b) System behaviour observation tasks c) Identifying scenarios from sample applications.	
2.	a) Create JMeter test plan b) Thread group setup c) Parameterized scripts d) Assertions and listeners	
3.	e) Execute load tests using GUI + CLI f) Server resource monitoring g) Generate and interpret HTML reports h) Identify and report issues.	
4.	e) SQL practice tasks f) CRUD and Join queries g) Validate tables, views and triggers h) Structural testing on sample schema.	
5.	a) Write SQL test validations b) Automate DB validation using JDBC c) Perform migration test validations d) Mini-project combining DB + performance checking	
	Total (P)	30 Periods
	Total (L+P)	7 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Understand performance testing fundamentals, terminologies, and methodologies.	L2 – Understand
CO2	Design, execute, and analyze performance test scripts using JMeter.	L4 – Analyze
CO3	Execute tests using both GUI and Non-GUI modes in JMeter and interpret HTML performance dashboards.	L3 – Apply
CO4	Apply SQL knowledge to validate database operations and schemas.	L3 – Apply
CO5	A Write SQL queries and use Java + JDBC framework for automated database testing	L4 – Analyze

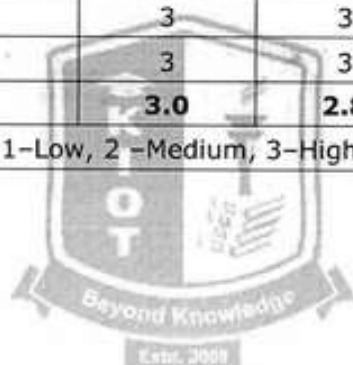
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	REFERENCE BOOKS:
1.	Elfriede Dustin, "Effective Software Testing: A Practical Guide", Pearson Education, 2nd Edition, 2018.
2.	Emily H. Halili, "Apache JMeter: Beginner's Guide", Packt Publishing, 3rd Edition, 2018.
3.	Rex Black, Erik van Veenendaal & Dorothy Graham, "Foundations of Software Testing: ISTQB Certification", Cengage Learning, 4th Edition, 2017.
4.	Thomas Connolly & Carolyn Begg, "Database Systems: A Practical Approach to Design, Implementation, and Management", Pearson Education, 6th Edition, 2015.
5.	Thomas Erl, "Database Design and Relational Theory", Pearson Education, 2nd Edition, 2019.

Mapping of COs with POs						
COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		3	2	3	1
CO2	2		3	3	3	2
CO3			3	3	3	1
CO4	2		3	3	3	1
CO5	2		3	3	3	2
Average	2.0		3.0	2.8	3.0	1.4
1-Low, 2 -Medium, 3-High						



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ME23IS501	ENVIRONMENTAL SAFETY		CP	L	T	P	C
			3	3	0	0	3
Type of Course	Open Elective						
Offering Dept.	Mechanical Engineering						
Programme & Branch	Common to M.E. (AE, PED and SE)					Version: 1.0	
Course Objectives:							
1.	To provide in depth knowledge in Principles of Environmental safety and its applications in various fields.						
2.	To give understanding of air and water pollution and their control.						
3.	To expose the students to the basis in hazardous waste management.						
4.	To provide knowledge on pollution monitoring and control devices.						
5.	To design emission measurement devices.						
UNIT-I	AIR POLLUTION					9	
	Classification and properties of air pollutants -Pollution sources - Effects of air pollutants on human beings, Animals, Plants, and Materials -Automobile pollution -Hazards of air pollution -Concept of clean coal combustion technology -Ultra violet radiation, infrared radiation, radiation from the sun-Hazards due to depletion of ozone-Deforestation, ozone holes, automobile exhausts, chemical factory stack emissions, CFC.						
UNIT-II	WATER POLLUTION					9	
	Classification of water pollutants-Health hazards-Sampling and analysis of water-Water treatment -Different industrial effluents and their treatment and disposal-Advanced wastewater treatment-Effluent quality standards and laws-Chemical industries, tannery, textile effluents -Common treatment.						
UNIT-III	HAZARDOUS WASTE MANAGEMENT					9	
	Hazardous waste management in India-Waste identification, characterization, and classification-Technological options for collection, treatment and disposal of hazardous waste, Selection charts for the treatment of different hazardous wastes-Methods of collection and disposal of solid wastes-Health hazards -Toxic and radioactive wastes-Incineration and vitrification-Hazards due to bio-process, dilution, standards, and restrictions-Recycling and reuse.						
UNIT-IV	ENVIRONMENTAL MEASUREMENT AND CONTROL					9	
	Sampling and analysis-Dust monitor -Gas analyzer- particle size analyzer-Lux meter- pH meter-Gas chromatograph-Atomic absorption spectrometer-Gravitational settling chambers, cyclone separators, scrubbers-Electrostatic precipitator, bag filter, maintenance-Control of gaseous emission by adsorption, absorption, and combustion methods-Pollution Control Board, laws.						

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UNIT-V	POLLUTION CONTROL IN PROCESS INDUSTRIES	9
	Pollution control in process industries -Cement, paper, petroleum, petroleum products, textile-Tanneries, thermal power plants-Dyeing and pigment industries -Eco-friendly energy.	
	Total(L)	45 Periods
	OPEN-ENDED PROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Illustrate and familiarize the basic concepts scope of environmental safety.	L2-Understand
CO2	Interpret the standards of professional conduct that are published by professional safety organizations and/or certification bodies.	L2- Understand
CO3	Explain the ways in which environmental health problems have arisen due to air and water pollution.	L2- Understand
CO4	Examine the role of hazardous waste management and use of critical thinking to identify and assess environmental health risks.	L4 - Analyze
CO5	Apply concepts of emission measurement and design emission measurement devices.	L3 - Apply
S.No.	REFERENCE BOOKS:	
1.	E C Wolfe, "Race to Save to Save Planet", Wadsworth Publishing Co., Belmont, 2006.	
2.	G T Miller, "Environmental Science: Working with the Earth", Wadsworth Publishing Co., Belmont, 11th Edition, 2006.	
3.	M J Hammer and M J Hammer, "Water and Wastewater Technology", Pearson Prentice Hall, 2006	
4.	Rao C S, "Environmental pollution Engineering", Wiley Eastern Limited, New Delhi, 2018.	
5.	S P Mahajan, "Pollution control in process industries", Tata McGraw Hill Publishing Company, New Delhi, 2006.	
6.	Varma and Braner, "Air pollution Equipment", Springer Publishers, Second Edition.	

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ME23IS502	ELECTRICAL SAFETY				CP	L	T	P	C
					3	3	0	0	3
Type of Course	Open Elective								
Offering Dept.	Mechanical Engineering								
Programme & Branch	Common to M.E. (AE, PED and SE)							Version: 1.0	
Course Objectives:									
1.	To impart knowledge on fundamental electrical concepts, equipment principles, and comply with safety regulations, including basic first aid.								
2.	To familiarize students with primary electrical hazards, insulation, and lightning protection measures.								
3.	To provide an in depth knowledge on functioning of fuses, circuit breakers, and safety measures against electrical faults.								
4.	To provide knowledge on equipment selection, safety features, and maintenance for electrical tools.								
5.	To familiarize students with hazardous zone classification, safe equipment, and safety measures in different environments.								
UNIT-I	CONCEPTS AND STATUTORY REQUIREMENTS							9	
	Introduction – electrostatics, electro magnetism, stored energy, energy radiation and electromagnetic interference– Working principles of electrical equipment-Indian electricity act and rules-statutory requirements from electrical inspectorate-international standards on electrical safety– first aid-Cardio Pulmonary Resuscitation(CPR).								
UNIT-II	ELECTRICAL HAZARDS							9	
	Primary and secondary hazards-shocks, burns, scalds, falls-human safety in the use of electricity. Energy leakage-clearances and insulation-classes of insulation-voltage classifications-excess energy current surges-Safety in handling of war equipment's-over current and short circuit current-heating effects of current-electromagnetic forces-corona effect-static electricity-definition, sources, hazardous conditions, control, electrical causes of fire and explosion-ionization, spark and arc ignition energy-national electrical safety code ANSI. Lightning, hazards, lightning arrestor, installation – earthing, specifications, earth resistance, earth pit maintenance.								
UNIT-III	PROTECTION SYSTEMS							9	
	Fuse, circuit breakers and overload relays– protection against over voltage and under voltage– safe limits of amperage – voltage –safe distance from lines-capacity and protection of conductor-joints-and connections, overload and short circuit protection-no load protection-earth fault protection. FRLS insulation-insulation and continuity test-system grounding-equipment grounding-earth leakage circuit breaker (ELCB) -cable wires-maintenance of ground-ground fault circuit interrupter-use of low voltage-electrical guards-Personal protective equipment-safety in handling hand held electrical appliances tools and medical equipment's.								

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UNIT-IV	SELECTION, INSTALLATION, OPERATION AND MAINTENANCE	9
	Role of environment in selection-safety aspects in application - protection and interlock-self diagnostic features and fail safe concepts-lock out and work permit system-discharge rod and earthing devices safety in the use of portable tools-cabling and cable joints-preventive maintenance.	
UNIT-V	HAZARDOUS ZONES	9
	Classification of hazardous zones-intrinsically safe and explosion proof electrical apparatus-increase safe equipment-their selection for different zones-temperature classification-grouping of gases-use of barriers and isolators-equipment certifying agencies.	
	Total(L)	45 Periods
	OPEN-ENDED PROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Demonstrate understanding of electrical concepts and legal compliance for safe operation, within regulatory constraints.	L2 - Understand
CO2	Identify and mitigate electrical hazards, ensuring safety adherence to protocols and guidelines.	L3 - Apply
CO3	Utilize protection systems effectively, ensuring electrical safety within specified standards.	L3 - Apply
CO4	Apply a safe and efficient process for selecting, installing, operating, and maintaining electrical equipment, adhering to industry regulations.	L3 - Apply
CO5	Develop expertise in managing hazardous zones safely, within the constraints of applicable safety standards.	L3 - Apply
S.No.	REFERENCE BOOKS:	
1.	" Accident prevention manual for industrial operations", N.S.C., Chicago, 1982.	
2.	Indian Electricity Act and Rules, Government of India.	
3.	Power Engineers – Handbook of TNEB, Chennai, 1989.	
4.	Martin Glov, "Electrostatic Hazards in powder handling", Research Studies Pvt. Ltd., England, 1988.	
5.	Fordham Cooper W, "Electrical Safety Engineering", Butterworth and Company, London, 1986.	

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ME23IS503	SAFETY IN ENGINEERING INDUSTRY				CP	L	T	P	C
					3	3	0	0	3
Type of Course	Open Elective								
Offering Dept.	Mechanical Engineering								
Programme & Branch	Common to M.E. (AE, PED and SE)							Version: 1.0	
Course Objectives:									
1.	To know the safety rules and regulations, standards and codes.								
2.	To study various mechanical machines and their safety importance.								
3.	To understand the principles of machine guarding and operation of protective devices.								
4.	To know the working principle of mechanical engineering processes such as metal forming and joining process and their safety risks.								
5.	To impart knowledge on finishing, inspection and testing operations in engineering industry.								
UNIT-I	SAFETY IN METAL WORKING MACHINERY AND WOOD WORKING MACHINES							9	
	General safety rules, principles, maintenance, Inspections of turning machines, boring machines, milling machine, planning machine and grinding machines, CNC machines. Wood working machinery, types, safety principles, electrical guards, work area, material handling, inspection, standards and codes- saws, types, hazards.								
UNIT-II	PRINCIPLES OF MACHINE GUARDING							9	
	Guarding during maintenance, Zero Mechanical State (ZMS), Definition, Policy for ZMS- guarding of hazards- point of operation protective devices, machine guarding, types, fixed guard, interlock guard, automatic guard, trip guard, electron eye, positional control guard, fixed guard fencing- guard construction- guard opening. Selection and suitability: lathe-drilling-boring-milling-grinding-shaping-sawing-shearing-presses-forge hammer -flywheels-shafts-couplings-gears-sprockets wheels and chains-pulleys and belts-authorized entry to hazardous installations-benefits of good guarding systems.								
UNIT-III	SAFETY IN WELDING AND GAS CUTTING							9	
	Gas welding and oxygen cutting, resistances welding, arc welding and cutting, common hazards, personal protective equipment, training, safety precautions in brazing, soldering and metalizing-explosive welding, selection, care and maintenance of the associated equipment and instruments - safety in generation, distribution and handling of industrial gases-colour coding-flashback arrestor- leak detection-pipe line safety-storage and handling of gas cylinders.								

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UNIT-IV	SAFETY IN COLD FARMING AND HOT WORKING OF METALS	9
	Cold working, power presses, point of operation safe guarding, auxiliary mechanisms, feeding and cutting mechanism, hand or foot-operated presses, power press electric controls, power press set up and die removal, inspection and maintenance - metal sheers-press brakes. Hot working safety in forging, hot rolling mill operation, safe guards in hot rolling mills – hot bending of pipes, hazards and control measures. Safety in gas furnace operation, cupola, crucibles, ovens-foundry health hazards, work environment, material handling in foundries, foundry production cleaning and finishing foundry processes.	
UNIT-V	SAFETY IN FINISHING, INSPECTION AND TESTING	9
	Heat treatment operations, electro plating, paint shops, sand and shot blasting, safety in inspection and testing, dynamic balancing, hydro testing, valves, boiler drums and headers, pressure vessels, air leak test, steam testing, safety in radiography, personal monitoring devices, radiation hazards, engineering and administrative controls, Indian Boilers Regulation. Health and welfare measures in engineering industry-pollution control in engineering industry- industrial waste disposal.	
	Total(L)	45 Periods
	OPEN-ENDED PROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Apply safety rules for maintaining and inspecting metal and wood working machines, ensuring industry standards.	L3 - Apply
CO2	Apply effective design strategies for machine guarding systems, emphasizing zero mechanical state (ZMS) during maintenance.	L3 - Apply
CO3	Demonstrate proficiency in safe welding and cutting, ensuring proper equipment selection, care, and maintenance.	L2 - Understand
CO4	Make use of safety measures in cold and hot metalworking, ensuring proper equipment setup, inspection, and maintenance.	L3 - Apply
CO5	Apply safety protocols in finishing, inspection, and testing, adhering to regulations and considering health and pollution control in engineering.	L3 - Apply
S.No.	REFERENCE BOOKS:	
1.	"Accident Prevention Manual", NSC, Chicago, 1982.	
2.	"Occupational safety Manual", BHEL, Trichy, 1988.	

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3.	John V. Grimaldi and Rollin H. Simonds, "Safety Management", All India Travelers Book seller, New Delhi, 1989.
4.	Krishnan N V, "Safety in Industry", Jaico Publishery House, 1996.
5.	Indian Boiler acts and Regulations, Government of India.
6.	"Safety in the use of wood working machines", HMSO, UK, 1992.
7.	"Health and Safety in welding and Allied processes", welding Institute, UK, High Tech. Publishing Ltd., London, 1989.



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ME23IS504	DESIGN OF EXPERIMENTS		CP	L	T	P	C
			3	3	0	0	3
Type of Course	Open Elective						
Offering Dept.	Mechanical Engineering						
Programme & Branch	Common to M.E. (AE, PED and SE)					Version: 1.0	
Course Objectives:							
1.	To impart knowledge on principles and steps in designing a statistically designed experiment.						
2.	To build foundation in analysing the data in single factor experiments and to perform post hoc tests.						
3.	To provide knowledge on analysing the data in factorial experiments.						
4.	To educate on analysing the data analysis in special experimental designs and Response Surface Methods.						
5.	To impart knowledge in designing and analysing the data in Taguchi's Design of Experiments to improve Process/Product quality.						
UNIT-I	EXPERIMENTAL DESIGN FUNDAMENTALS					9	
	Importance of experiments, experimental strategies, basic principles of design, terminology, ANOVA, steps in experimentation, sample size, normal probability plot, linear regression models.						
UNIT-II	SINGLE FACTOR EXPERIMENTS					9	
	Completely randomized design, Randomized block design, Latin square design, Statistical analysis, estimation of model parameters, model adequacy checking, pair wise comparison tests.						
UNIT-III	MULTIFACTOR EXPERIMENTS					9	
	Two and three factor full factorial experiments, Randomized block factorial design, Experiments with random factors, rules for expected mean squares, approximate F-tests. 2 ^K factorial Experiments.						
UNIT-IV	SPECIAL EXPERIMENTAL DESIGNS					9	
	Blocking and confounding in 2 ^K designs. Two level Fractional factorial design, nested designs, Split plot design, Introduction to Response Surface Methods.						
UNIT-V	TAGUCHI METHODS					9	
	Steps in experimentation, design using Orthogonal Arrays, data analysis, Robust design,- control and noise factors, S/N ratios, parameter design, Multi-level experiments, Multi-response optimization, Introduction to Shainin DOE.						
	Total(L)					45 Periods	

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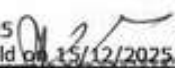
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	OPEN-ENDED PROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
C01	Interpret the Design of Experiments principles, strategizing experiment design within practical resource considerations and goals.	L2 - Understand
C02	Analyze single-factor experiment data, focusing on randomization and pair-wise comparison tests.	L4 - Analyze
C03	Analyze multifactor experiment data, applying rules for expected mean squares and approximate F-tests.	L4 - Analyze
C04	Apply special experimental designs, minimize confounding effects, optimize data collection, and introduce Response Surface Methods with practical considerations.	L3 - Apply
C05	Apply Taguchi-based approaches for quality evaluation, emphasizing practical experimentation with orthogonal arrays and multi-response optimization.	L3 - Apply
S.No.	REFERENCE BOOKS:	
1.	Krishnaiah K and Shahabudeen P, "Applied Design of Experiments and Taguchi Methods", PHI learning private Ltd., 2012.	
2.	Montgomery D C, "Design and Analysis of Experiments", John Wiley and Sons, Eighth edition, 2012.	
3.	Nicolo Belavendram, "Quality by Design: Taguchi techniques for industrial Experimentation", Prentice Hall, 1995.	
4.	Phillip J Rose, "Taguchi techniques for quality Engineering", McGraw Hill, 1996.	


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ME23IS505	CIRCULAR ECONOMY	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	Mechanical Engineering					
Programme & Branch	Common to M.E. (AE, PED and SE)					Version: 1.0
Course Objectives:						
1.	To equip graduates with circularity expertise for diverse national and international job opportunities.					
2.	To develop skilled manpower and foster entrepreneurship in Circular Economy.					
3.	To facilitate student-professional interactions for real-world exposure in technology, research, innovation, and circular business models.					
4.	To inspire students to address circularity business needs and pursue Research and Development (R&D) and entrepreneurship.					
5.	To cultivate environmentally conscious entrepreneurs through core competencies in environmental education and collaborative university-industry partnerships.					
UNIT-I	INTRODUCTION TO CIRCULAR ECONOMY					9
	Linear Economy and its emergence, Economic and Ecological disadvantages of linear economy, Replacing Linear economy by Circular Economy, Development of Concept of Circular Economy, A differential - Linear Vs Circular Economy.					
UNIT-II	CHARACTERISTICS OF CIRCULAR ECONOMY					9
	Material recovery, Waste Reduction, reducing negative externalities, Explaining Butterfly diagram, Concept of Loops.					
UNIT-III	CIRCULAR DESIGN, INNOVATION AND ASSESSMENT					9
	Zero waste: Waste Management in context of Circular Economy, Circular design, Research and innovation, LCA, Circular Business.					
UNIT-IV	CASE STUDIES					9
	Business models, Solid Waste Management / Wastewater, Plastics: A case study, EPR: polluters pay principle, Industrial symbiosis/ Eco-parks.					
UNIT-V	LEGAL AND POLICY FRAMEWORK					9
	Role of governments and networks, Sharing best practices, Universal circular economy policy goals, India and CE strategy, ESG.					
	Total(L)					45 Periods
	OPEN-ENDED PROBLEMS/QUESTIONS					
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.					

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Differentiate Circular Economy from Linear Economy and showcase its practical application.	L2 - Understand
CO2	Apply Circular Economy principles, incorporating material recovery and waste reduction to illustrate the Butterfly diagram and emphasize the loops within the circular system.	L3 - Apply
CO3	Apply circular design and innovation principles, assess sustainability in Circular Economy, and examine circular business models	L3 - Apply
CO4	Analyze case studies on circular economy from different fields and connect these cases to Circular Economy concepts professionally.	L4 - Analyze
CO5	Infer government roles, share best practices, and articulate Circular Economy policy goals, demonstrating expertise in legal frameworks with an ESG focus, especially in India.	L2 - Understand
S.No.	REFERENCE BOOKS:	
1.	María-Laura Franco-García, Jorge Carlos Carpio-Aguilar, Hans Bressers, "Towards Zero Waste: Circular Economy Boost, Waste to Resources", Springer International Publishing, 2019.	
2.	Marcello Tonelli and Nicolo Cristoni, "Strategic Management and the Circular Economy", Routledge, 2018.	
3.	Sadhan Kumar Ghosh, "Circular Economy: Global Perspective", Springer, 2020.	
4.	Stahel and Walter R, "The Circular Economy: A User's Guide", Routledge, 2019.	
5.	Lerwen Liu, Seeram Ramakrishna, "An Introduction to Circular Economy", Springer Singapore, 2021.	

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M.E. / M.Tech. / M.C.A. Regulations 2023

ME23AE501	FUNDAMENTALS OF AUTOMOTIVE ELECTRONICS	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	M.E. AUTOMOTIVE ELECTRONICS					
Programme & Branch	Common to M.E (SE, PED, ISE)				Version: 1.0	
Course Objectives:						
1.	To learn the basics of automotive electronics and system architecture.					
2.	To understand different sensors, actuators, and control methods in vehicles.					
3.	To study engine and powertrain electronics, including emission and hybrid systems.					
4.	To learn about automotive communication networks and diagnostic systems.					
5.	To explore safety systems, comfort features, and new technologies in automobiles.					
UNIT-I	INTRODUCTION TO AUTOMOTIVE ELECTRONICS				9	
	Evolution of automotive Electronics-Modern automotive electrical system Overview-Vehicle architecture: Body electronics- Powertrain electronics- Chassis electronics-Automotive sensors and actuators; characteristics, Applications -Introduction to ECUs: architecture- microcontrollers-memory-Automotive Wiring Harness- Connectors- Grounding-Shielding- Overview of automotive standards: ISO-SAE-AIS.					
UNIT-II	SENSORS, ACTUATORS & CONTROL SYSTEMS				9	
	Position sensors: TPS-Hall sensors-crank/cam Sensors-Speed sensors-pressure sensors-temperature Sensors-Oxygen/ λ sensor-MAF-MAP-knock Sensor-Actuators: fuel injectors-ignition coils, stepper motors, BLDC motors, relays. Engine control system basics: closed-loop control, feedback, PID-Electronic throttle control system-Signal conditioning and data acquisition in vehicles.					
UNIT-III	POWERTRAIN AND ENGINE MANAGEMENT ELECTRONICS				9	
	Electronic Engine Control (EEC) architecture-Fuel injection systems: MPFI, GDI, CRDI-Ignition systems: electronic ignition, spark timing control-Air-Air-fuel ratio control-emission control electronics-Exhaust after-treatment systems (EGR, SCR, DPF sensors & control)-Transmission control electronics: TCU, sensors & actuation-Hybrid and electric vehicle basics: motor controllers, inverters, battery electronics.					
UNIT-IV	VEHICLE COMMUNICATION NETWORKS				9	
	Automotive communication basics: need for networking protocol: frame structure- arbitration- error handling-LIN- Flex Ray- MOST - applications and differences- Automotive Ethernet --Network architecture in modern vehicles.					
UNIT- V	SAFETY, COMFORT SYSTEMS				9	
	Electronic stability control (ESC)- traction control (TCS)-Anti-Lock Braking System (ABS): sensors- electronic control-Airbag					

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	systems: crash sensors- deployment logic-Body control electronics: lighting- windows- HVAC- Immobilizers-ADAS overview: radar- camera- ultrasonic systems-Cybersecurity challenges in automotive electronics-Emerging trends: Software defined vehicles-Autonomous driving electronics.	
	Total (L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course the students will be able to:	BLOOM'S Taxonomy
CO1	Explain the basic structure and components of automotive electronic systems.	L2 - Understand
CO2	Identify common sensors, actuators, and their functions in vehicles.	L2 - Understand
CO3	Describe electronic control of engine, fuel, and emission systems.	L2 - Understand
CO4	Understand communication networks and diagnostic procedures in vehicles.	L2 - Understand
CO5	Outline important safety features and modern trends in automotive electronics.	L2 - Understand
	REFERENCE BOOKS:	
1.	William B. Ribbens, "Understanding Automotive Electronics", Newnes, 8 th Edition, 2017.	
2.	Tom Denton, "Automotive Electronics", Routledge, 3 rd Edition, 2021.	
3.	Robert Bosch GmbH, "Bosch Automotive Handbook", Bentley Publishers, 10 th Edition, 2018.	
4.	Robert Bosch GmbH, "Bosch Automotive Electrics and Automotive Electronics: Systems and Components, Networking and Hybrid Drive", Springer, 6 th Edition, 2014.	
5.	James D. Halderman, "Automotive Electricity and Electronics", Pearson, 5 th Edition, 2016.	

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M.E./M.Tech./M.C.A. Regulations 2023

ME23AE502	VEHICLE SENSORS AND ELECTRONIC CONTROL SYSTEMS	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	M.E. AUTOMOTIVE ELECTRONICS					
Programme & Branch	Common to M.E (SE, PED, ISE)				Version: 1.0	
Course Objectives:						
1.	To understand the role, types, and characteristics of sensors used in modern vehicles.					
2.	To learn powertrain sensors and their use in engine and transmission control.					
3.	To study chassis, dynamics, and safety-related sensors and their integration.					
4.	To explore body electronics sensors used for comfort, environment, and security systems.					
5.	To gain knowledge of ADAS sensors, sensor fusion, diagnostics, safety, and emerging technologies.					
UNIT-I	INTRODUCTION TO VEHICLE SENSORS & MEASUREMENT TECHNOLOGY					9
	Need for sensors in automotive systems-Classification of sensors: mechanical- electrical- magnetic- thermal- optical- chemical-Sensor characteristics: sensitivity- accuracy- drift- hysteresis- linearity- bandwidth-Signal conditioning: filtering- amplification- analog-digital conversion-Smart sensors and micro-electromechanical systems (MEMS)-Automotive standards: ISO- SAE- AEC-Q100 for sensor qualification.					
UNIT-II	POWERTRAIN SENSORS AND ELECTRONIC CONTROL					9
	Engine sensors: -Crankshaft & camshaft position Sensors-Manifold Absolute Pressure (MAP) Sensor-Mass Air Flow (MAF) Sensor-Throttle Position Sensor (TPS)-Knock sensor- oxygen (λ) Sensor-Coolant temperature and intake air temperature Sensors-Fuel injection- ignition timing- and combustion control using Sensors-Transmission sensors: speed- gear-position-clutch pressure sensors--Electronic Engine Management System (EEMS) Architecture-Closed-loop and open-loop control strategies.					
UNIT-III	CHASSIS, DYNAMICS & SAFETY SENSORS					9
	Wheel speed sensors for ABS-Yaw rate- lateral acceleration- and gyro Sensors-Steering angle sensors and suspension height Sensors-Brake pressure sensors- tire pressure monitoring systems (TPMS)-Airbag sensors: crash accelerometers- impact detection Systems-Electronic Stability Control (ESC)-Traction Control System (TCS) sensors and integration.					
UNIT-IV	BODY ELECTRONICS, ENVIRONMENT & COMFORT SYSTEM SENSORS					9
	Climate control sensors: humidity- cabin temperature- sunlight sensors-Rain sensors- wiper control- interior air quality sensors-Parking sensors: ultrasonic- radar-based-Interior sensors: occupant detection- seat-belt sensors- weight classification sensors-Door- window- anti-pinch and proximity sensors-Immobilizer and smart key system sensors-Comfort system control units and communication interfaces.					

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UNIT- V	ADVANCED DRIVER ASSISTANCE SYSTEM (ADAS) & ELECTRONIC CONTROL	9
	ADAS sensor technologies: - Radar- LiDAR- cameras- ultrasonic sensors-Sensor fusion and perception systems-Electronic control units for ADAS-Autonomous vehicle sensing architecture- Diagnostics: OBD-II- UDS- fault codes- sensor health monitoring-Functional safety: ISO 26262 considerations for sensor integration-Cybersecurity for sensor networks-Future trends: solid-state LiDAR- high-resolution radar- V2X sensors- self-calibrating sensors.	
	Total (L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course the students will be able to:	BLOOM'S Taxonomy
C01	Explain different vehicle sensors and their basic characteristics.	L2 - Understand
C02	Describe powertrain sensors and their role in fuel, ignition, and transmission control.	L2 - Understand
C03	Interpret sensor functions for chassis control, vehicle dynamics, and safety systems.	L2 - Understand
C04	Identify body electronics sensors used for comfort, environment, and occupant safety.	L3 - Apply
C05	Evaluate ADAS sensor technologies, diagnostic methods, safety standards, and future trends.	L3 - Apply
	REFERENCE BOOKS:	
1.	Tan Sinclair, "Sensors and Transducers", Newnes, 3rd Edition, 2001.	
2.	Konrad Reif (Ed.), "Automotive Sensors", Springer, 1st Edition, 2014.	
3.	William B. Ribbens, "Understanding Automotive Electronics", Newnes, 8th Edition, 2017.	
4.	Robert Bosch GmbH, "Bosch Automotive Handbook", Bentley Publishers, 10th Edition, 2018.	
5.	Dale R. Patrick and Stephen W. Fardo, "Understanding Sensors", Prentice Hall, 1999.	

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ME23AE503	AUTOMOTIVE EMBEDDED SYSTEMS		CP	L	T	P	C
			3	3	0	0	3
Type of Course	Open Elective						
Offering Dept.	M.E. AUTOMOTIVE ELECTRONICS						
Programme & Branch	Common to M.E (SE, PED, ISE)					Version: 1.0	
Course Objectives:							
1.	To understand the fundamentals, architecture, and standards of embedded systems used in modern vehicles.						
2.	To study automotive sensors, actuators, and interfacing techniques for real-time control.						
3.	To learn ECU hardware/software architecture, AUTOSAR framework, and functional safety concepts.						
4.	To explore automotive communication networks, diagnostic protocols, and network security.						
5.	To analyze embedded systems used in advanced vehicle functions, EVs, ADAS, and emerging trends.						
UNIT-I	INTRODUCTION TO EMBEDDED SYSTEMS IN AUTOMOBILES					9	
	Overview of embedded systems: definition- characteristics- real-time constraints-Evolution and role of embedded electronics in modern vehicles-Embedded system architecture: hardware- firmware- software stack-Microcontrollers for automotive applications: ARM- Infineon Aurix- Renesas- NXP- Memory systems: Flash- EEPROM- RAM- external memory- Introduction to real-time operating systems (RTOS)-Overview of automotive standards (ISO- SAE- AUTOSAR- AEC-Q100).						
UNIT-II	AUTOMOTIVE SENSORS, ACTUATORS & INTERFACING					9	
	Sensors: position- speed- pressure- temperature- oxygen- radar- Camera- Actuators: motors- solenoids- injectors- relays- Valves- Signal conditioning & sensor interfaces-A/D- D/A converters- PWM- timers and Counters-Hardware-software interfacing- GPIO-Diagnostics and fault detection at sensor/actuator Level-Case study: Electronic Throttle Control Module.						
UNIT-III	ELECTRONIC CONTROL UNITS (ECUS) AND SOFTWARE ARCHITECTURE					9	
	ECU architecture: CPU- peripherals- communication modules- power supply-Types of ECUs: Engine Control Unit- TCU- BCU- ABS/ESC- ADAS ECUs-Embedded software development lifecycle: V-model- ASPICE-Bootloaders and ECU flashing- Introduction to AUTOSAR architecture: - Classic Platform: BSW- RTE- SWC-Adaptive Platform: POSIX- middleware- services-Model-based development (MATLAB/Simulink)- Functional safety concepts: ISO 26262- HARA- safety goals- ASIL levels.						
UNIT-IV	AUTOMOTIVE COMMUNICATION NETWORKS & PROTOCOLS					9	
	Need for in-vehicle networking-CAN: frame formats- arbitration- error handling- CAN FD-LIN: master-slave concept- scheduling-FlexRay: time-triggered communication-MOST and Automotive Ethernet-UDS diagnostics- OBD-II- DTCs-Gateway						

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76

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	ECUs and network security-Time-synchronized networking and deterministic communication.	
UNIT- V	EMBEDDED SYSTEMS FOR ADVANCED VEHICLE FUNCTIONS	9
	Powertrain embedded systems: engine control- transmission control-Chassis embedded systems: ABS- ESC- TPMS- steering systems-Body electronics: lighting- HVAC- seat control-keyless entry-ADAS control systems: path planning-perception- sensor fusion basics-Embedded systems in Electric Vehicles (EV): - BMS- motor controllers- DC-DC converters-On-board chargers- inverter control-Embedded system testing: HiL- SiL- MiL-Cybersecurity in automotive embedded systems.	
	Total (L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course the students will be able to:	BLOOM'S Taxonomy
CO1	Explain the structure, features, and standards of automotive embedded systems.	L2 - Understand
CO2	Describe the operation of sensors, actuators, and interface modules used in vehicles.	L2 - Understand
CO3	Interpret ECU architectures, AUTOSAR platforms, safety methods, and software development processes.	L2 - Understand
CO4	Understand vehicle communication protocols, diagnostics, and network protection mechanisms.	L2 - Understand
CO5	Evaluate embedded systems applied to powertrain, chassis, EVs, ADAS, and future vehicle technologies.	L3 - Apply
	REFERENCE BOOKS:	
1.	Robert Bosch GmbH, "Automotive Electrics and Automotive Electronics", Springer, 6th Edition, 2014.	
2.	Nicolas Navet & François Simonot-Lion, "Automotive Embedded Systems Handbook", CRC Press, 1st Edition, 2008.	
3.	William B. Ribbens, "Understanding Automotive Electronics", Newnes, 8th Edition, 2017.	
4.	Matthias Klauda, "AUTOSAR – A Practical Approach", Springer, 1st Edition, 2020.	
5.	John Wiley & Sons, "Embedded Systems Handbook", CRC Press, 2nd Edition, 2009.	

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M.E./M.Tech/M.C.A. Regulations 2023

ME23AE504	AUTOMOTIVE COMMUNICATION NETWORKS	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	M.E. AUTOMOTIVE ELECTRONICS					
Programme & Branch	Common to M.E (SE, PED, ISE)				Version: 1.0	
Course Objectives:						
1.	To understand the need, evolution, and architecture of in-vehicle communication networks.					
2.	To study CAN and CAN FD protocols, frame formats, error handling, and applications.					
3.	To learn LIN, FlexRay, MOST networks and their roles in vehicle communication.					
4.	To explore automotive Ethernet, diagnostic protocols, and ECU flashing methods.					
5.	To analyze network integration, security mechanisms, testing approaches, and future trends					
UNIT-I	INTRODUCTION TO IN-VEHICLE NETWORKING				9	
	Need for networking in modern vehicles-Evolution of automotive electronics and distributed Communication requirements: speed- latency- reliability- Determinism- Network topologies: bus- star- ring- hybrid-OSI model and relevance to automotive Networks-Classification of automotive networks: low-speed- high-speed- multimedia- Safety-Critical- Overview of standards: ISO- SAE- AEC- AUTOSAR networking concepts.					
UNIT-II	CAN AND CAN FD NETWORKS				9	
	Controller Area Network (CAN): features- architecture-CAN frame structure: base/extended frames- ID- DLC- CRC-Bit stuffing- arbitration- error detection & handling-CAN protocols: Classical CAN vs CAN FD-Physical layer requirements and Transceivers-Higher-layer protocols: CANopen- J1939- XCP- Use cases: powertrain- chassis- body systems-CAN bus diagnostics and troubleshooting.					
UNIT-III	LIN, FLEXRAY AND MOST NETWORKS				9	
	LIN (Local Interconnect Network)-Master-slave architecture-Frames- schedule tables- clusters-Applications: body electronics- door modules- HVAC- FlexRay- Motivation for FlexRay: high-speed- deterministic communication-Static and dynamic segments- TDMA concept-Fault tolerance and redundancy-Applications: safety-critical (braking- steering- chassis)-MOST (Media Oriented Systems Transport)-Multimedia & infotainment communication-Network layers- synchronous/asynchronous channels-Applications: audio- video- telematics.					
UNIT-IV	AUTOMOTIVE ETHERNET & DIAGNOSTICS COMMUNICATION				9	
	Automotive Ethernet-Introduction and need for high-bandwidth communication-100BASE-T1/1000BASE-T1 automotive standards -TSN (Time-Sensitive Networking) for deterministic performance-Applications: ADAS, autonomous					

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	driving, camera & sensor networks - Diagnostics & Flashing- On-board diagnostics (OBD-II): modes- PIDs-Unified Diagnostic Services (UDS): request/response- service IDs- KWP2000 basics-ECU flashing and reprogramming methods- DTCs (Diagnostic Trouble Codes) and fault memory.	
UNIT- V	NETWORK INTEGRATION, SECURITY	9
	Network management: sleep/wake-up- bus-off recovery- Gateway ECUs: routing- filtering- protocol translation- AUTOSAR communication stack: COM- PDUR- CANIF- ETHIF- Cybersecurity for in-vehicle networks: -Threats- attack surfaces-Secure Onboard Communication (SecOC)-Intrusion detection systems - Testing and validation: HiL- SiL- bus analyzers- network simulators.	
	Total (L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course the students will be able to:	BLOOM'S Taxonomy
CO1	Explain the purpose, architecture, and types of automotive communication networks.	L2 - Understand
CO2	Describe CAN, CAN FD protocols, frame structure, and their use in vehicle subsystems.	L2 - Understand
CO3	Interpret communication principles of LIN, FlexRay, and MOST networks and their applications.	L3 - Apply
CO4	Understand automotive Ethernet, diagnostic services, and ECU programming techniques.	L3 - Apply
CO5	Apply basic network integration ideas and identify common security issues and new trends.	L3 - Apply
	REFERENCE BOOKS:	
1.	Kirsten Matheus and Thomas Königseder, "Automotive Ethernet", Springer, 2nd Edition, 2021.	
2.	Pawel Gaj and Josep Aguilar, "In-Vehicle Communication Networks", Wiley, 1st Edition, 2016.	
3.	Konrad Reif (Ed.), "Automotive Mechatronics: Automotive Networking, Driving Stability Systems, Electronics", Springer, 2nd Edition, 2015.	
4.	P.R. Kumar & P.R. Sundararajan, "Automotive Communication Protocols", McGraw Hill, 1st Edition, 2019.	
5.	Florian Schäfer, "CAN System Engineering: From Theory to Practical Applications", Springer, 1st Edition, 2016.	

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M.E./M.Tech/M.C.A. Regulations 2023

ME23AE505	INTRODUCTION TO ELECTRIC AND HYBRID VEHICLE ELECTRONICS	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	M.E. AUTOMOTIVE ELECTRONICS					
Programme & Branch	Common to M.E (SE, PED, ISE)				Version: 1.0	
Course Objectives:						
1.	To provide basic knowledge of electric and hybrid vehicle types and architectures.					
2.	To explain electric motors, power electronic converters and control techniques used in EVs.					
3.	To describe battery technologies, BMS functions and energy management systems.					
4.	To illustrate hybrid powertrain configurations and control strategies.					
5.	To introduce EV control systems, communication, safety and future technologies.					
UNIT-I	FUNDAMENTALS OF ELECTRIC & HYBRID VEHICLES				9	
	Evolution of vehicle electrification-Classification of EVs: HEV, PHEV, BEV, FCEV-Vehicle system architecture: conventional vs hybrid vs electric-Energy flow in hybrid and electric powertrains-Drive cycles: NEDC, WLTP, FTP-Overview of EV subsystems: traction motor, inverter, battery pack, charger-Automotive standards & regulations related to EVs (ISO, AIS, UN ECE).					
UNIT-II	ELECTRIC MACHINES & POWER ELECTRONICS				9	
	Types of electric motors for EVs: BLDC, PMSM, Induction Motor, Switched Reluctance Motor-Torque-speed characteristics & efficiency maps-Motor control techniques: FOC, DTC, PWM-Power electronics converters: - DC-DC (buck, boost, bidirectional)-Inverters (single-phase, three-phase)-On-board chargers (AC/DC)-Power semiconductor devices: MOSFET, IGBT, SiC, GaN -Protection circuits, EMI/EMC considerations.					
UNIT-III	BATTERIES, BMS & ENERGY MANAGEMENT				9	
	Battery basics: electrochemistry, Li-ion chemistries (NMC, LFP, NCA), temperature effects-Battery pack design: cell configuration, thermal management-Battery parameters: SOC, SOH, SOP, SOF-Battery Management System (BMS) architecture: - Centralized, distributed, modular topologies-Cell balancing: passive and active-Charging methods: CC-CV, fast charging, regenerative charging-Charging standards: CCS, CHAdeMO, GB/T, IEC 61851-Supercapacitors and hybrid energy storage systems.					
UNIT-IV	HYBRID POWERTRAIN ARCHITECTURE & CONTROL				9	
	Hybrid configurations: - Series, parallel, series-parallel, power-split hybrids-E-Drive unit (EDU) and e-Axle design-Transmission systems for EVs-Energy management strategies:-Rule-based, optimization-based-Torque split and power flow control-Regenerative braking system: electronics.					

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	control & safety-Thermal management for powertrain electronics	
UNIT- V	ELECTRONIC CONTROL SYSTEMS, COMMUNICATION & FUTURE TRENDS	9
	Vehicle Control Unit (VCU) architecture-Powertrain ECUs: inverter controller, motor controller, BMS controller-Vehicle communication networks: CAN, CAN FD, LIN, Automotive Ethernet-Diagnostics: OBD for EV/HEV, UDS, DTC handling-Functional safety: ISO 26262 for EV systems-Cybersecurity in EV electronics-Emerging technologies: - Solid-state batteries-Wireless charging systems-V2G/V2H/V2X integration-Software-defined EV architectures-AI-based energy management-Autonomous/ADAS considerations for EVs	
	Total (L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course the students will be able to:	BLOOM'S Taxonomy
CO1	Identify different types of electric and hybrid vehicles and their subsystems.	L2 - Understand
CO2	Explain motor types, converters and basic control methods used in EVs.	L2 - Understand
CO3	Describe batteries, BMS functions, charging methods and storage systems.	L3 - Apply
CO4	Design hybrid architectures and summarize energy control strategies.	L3 - Apply
CO5	Apply basic EV control and diagnostic concepts in real systems.	L3 - Apply
	REFERENCE BOOKS:	
1.	Iqbal Husain, "Electric and Hybrid Vehicles: Design Fundamentals", CRC Press, 3rd Edition, 2021.	
2.	James Larminie & John Lowry, "Electric Vehicle Technology Explained", Wiley, 2nd Edition, 2012.	
3.	Sandeep Dhameja, "Electric Vehicle Battery Systems", Newnes, 1st Edition, 2001	
4.	S. M. Mousazadeh, "Hybrid Electric Vehicles, Springer", 1st Edition, 2023.	
5.	Gonzalez & Ali Emadi, "Advanced Electric Drive Vehicles", CRC Press, 1st Edition, 2017.	

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ME23PE501	ELECTRIC VEHICLE TECHNOLOGY	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	M.E. – POWER ELECTRONICS AND DRIVES					
Programme & Branch	Common to M.E (ISE, AE and SE)				Version: 1.0	
Course Objectives:						
1.	To explain the historical evolution and the need for electric and hybrid vehicles.					
2.	To describe electric vehicle architectures, powertrain layouts, and performance characteristics.					
3.	To understand energy storage technologies, battery characteristics, and charging systems.					
4.	To analyze different electric motors, power electronics drives, and control strategies used in EVs.					
5.	To understand EV performance parameters, safety aspects, and major challenges.					
UNIT-I	NEED FOR ELECTRIC VEHICLES					9
	History and Need for Electric and Hybrid Vehicles - Social and Environmental Importance of Hybrid and Electric Vehicles - Impact of Modern drive-trains on Energy Supplies - Comparison of Diesel, Petrol, Electric and Hybrid Vehicles – Limitations and Technical Challenges					
UNIT-II	ELECTRIC VEHICLE ARCHITECHTURE					9
	Electric Vehicle Types and Layout – Traction Motor Characteristics - Transmission Requirements - Concepts of Hybrid Electric Drive Train - Architecture of Series and Parallel Hybrid Electric Drive Train - Mild and Full Hybrids - Plug-in Hybrid Electric Vehicles.					
UNIT-III	ENERGY STORAGE					9
	Batteries – Types: Lead Acid Batteries, Nickel based Batteries, and lithium based batteries - Electrochemical Reactions - Energy Efficiency - Battery Charging Methods – Charger Types - Battery Cooling Methods.					
UNIT – IV	ELECTRIC DRIVES AND CONTROL					9
	Types of Electric Motors – Advantages and Limitations - DC Motor Drives and Control - Induction Motor Drives and Control - PMSM and Brushless DC motor - Speed Controllers.					
UNIT – V	EV – PERFORMANCE AND CHALLENGES					9
	Range, Efficiency, Acceleration, Gradeability, and Braking - Safety Aspects in Electric Vehicles – Challenges in Electric Vehicle Charging Infrastructure, Recycling, and Environmental Impact.					
	Total(L)					45 Periods
	* INTERNAL EVALUATION ONLY					
	OPEN-ENDED PROBLEMS / QUESTIONS					
	Course specific Open Ended Problems will be solved during classroom teaching. Such problems can be given as Assignment.					CHAIRPERSON

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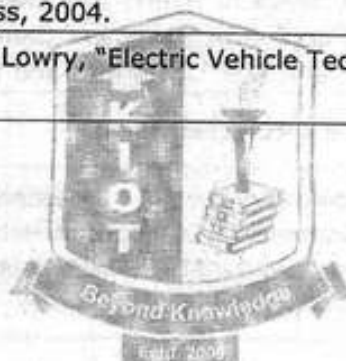
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	evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Understand the need for electric vehicles, their benefits, and limitations compared to conventional vehicles.	L2 - Understand
CO2	Explain electric vehicle architecture, types, and hybrid drive-train concepts.	L2 - Understand
CO3	Understand different energy storage systems, battery operation, and charging methods.	L2 - Understand
CO4	Identify electric motors used in EVs and explain their control techniques.	L2 - Understand
CO5	Apply EV performance parameters techniques to solve safety issues, and key implementation challenges.	L3 - Apply
	REFERENCE BOOKS:	
1.	Iqbal Hussein, "Electric and Hybrid Vehicles: Design Fundamentals", 2 nd Edition CRC Press, 2011.	
2.	Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, Ali Emadi, "Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design", CRC Press, 2004.	
3.	James Larminie, John Lowry, "Electric Vehicle Technology Explained", Wiley, 2003.	



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ME23PE502	ENERGY CONSERVATION AND MANAGEMENT	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	M.E. – POWER ELECTRONICS AND DRIVES					
Programme & Branch	Common to M.E (ISE, AE and SE)	Version: 1.0				
Course Objectives:						
1.	To learn the present energy scenario and the need for energy conservation.					
2.	To learn basics of refrigeration and air conditioning and methods to improve energy efficiency.					
3.	To understand efficient lighting systems, sensors, and energy saving measures in appliances.					
4.	To learn concepts of energy-efficient buildings, passive design, and renewable energy use.					
5.	To understand the need, types, working, and applications of energy storage systems.					
UNIT-I	ENERGY SCENARIO					9
	Primary Energy Resources - Sectorial Energy Consumption (Domestic, Industrial and other sectors) - Energy Pricing - Energy Conservation and its Importance - Energy Conservation Act-2001 and its features - Energy Star Rating.					
UNIT-II	HVAC SYSTEMS					9
	Basics of Refrigeration and Air Conditioning - COP / EER / SEC Evaluation - SPV System Design & Optimization for Solar Refrigeration.					
UNIT-III	LIGHTING AND APPLIANCES					9
	Specification of Luminaries - Types - Efficacy - Selection & Application - Time Sensors - Occupancy Sensors - Energy Conservation Measures in Electronic Devices.					
UNIT - IV	ENERGY EFFICIENT BUILDINGS					9
	Conventional versus Energy Efficient Buildings - Landscape Design - Envelope Heat Loss and Heat Gain - Passive Cooling and Heating - Renewable Sources Integration.					
UNIT - V	ENERGY STORAGE TECHNOLOGIES					9
	Necessity & Types of Energy Storage - Thermal Energy Storage - Battery Energy Storage, Charging and Discharging - Hydrogen Energy Storage & Super Capacitors - Energy Density and Safety Issues - Applications.					
	Total (L)					45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS					
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.					

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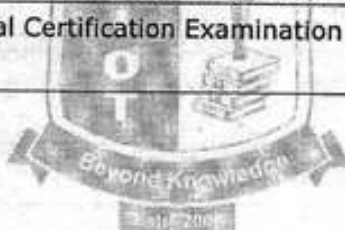
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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Understand technical aspects of energy conservation scenario.	L2 - Understand
CO2	Understand the basics of refrigeration and HVAC systems and how they help in saving energy.	L2 - Understand
CO3	Learn how energy-efficient lighting and electronic appliances reduce power consumption.	L2 - Understand
CO4	Apply concepts of energy-efficient and sustainable building design.	L3 - Apply
CO5	Gain basic knowledge of different energy storage technologies and their applications.	L2 - Understand
	REFERENCE BOOKS:	
1.	Yogi Goswami, Frank Kreith, "Energy Efficiency and Renewable energy Handbook", CRC Press, 2016.	
2.	ASHRAE Handbook 2020 – "HVAC Systems & Equipment"	
3.	Paolo Bertoldi, Andrea Ricci, Anibal de Almeida, "Energy Efficiency in Household Appliances and Lighting", Springer, 2001.	
4.	David A. Bainbridge, Ken Haggard, Kenneth L. Haggard, "Passive Solar Architecture: Heating, Cooling, Ventilation, Daylighting, and More using Natural Flows", Chelsea Green Publishing, 2011.	
5.	Guide book for National Certification Examination for Energy Managers and Energy Auditors.	



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ME23PE503	SMART GRID	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	M.E. – POWER ELECTRONICS AND DRIVES					
Programme & Branch	Common to M.E (ISE, AE and SE)					Version: 1.0
Course Objectives:						
1.	To study about smart grid technologies, different smart meters and advanced metering infrastructure.					
2.	To know about the function of smart grid.					
3.	To familiarize the power quality management issues in smart grid.					
4.	To familiarize the high performance computing for smart grid applications					
5.	To get familiarized with the communication networks for smart grid applications					
UNIT-I	INTRODUCTION					9
	Evolution of Electric Grid - Concept, Definitions and Need for Smart Grid - Smart Grid Drivers, functions, Opportunities, Challenges and benefits - Difference between Conventional & Smart Grid - Smart Grid Initiative in India - Case Study.					
UNIT-II	SMART GRID TECHNOLOGIES					9
	Integration of Renewable Energy Sources - Smart substations, Substation Automation - Protection and control - Fault Detection, Isolation and service restoration - Outage management - Grid to Vehicle and Vehicle to Grid charging concepts.					
UNIT-III	ADVANCED METERING INFRASTRUCTURE					9
	Introduction to Smart Meters - Advanced Metering infrastructure (AMI) drivers and benefits - AMI protocols and standards - AMI needs in the Smart Grid - Demand side Management and Demand Response Programs, Demand Pricing and Time of Use, Real Time Pricing and Peak Time Pricing.					
UNIT - IV	POWER QUALITY MANAGEMENT					9
	Power Quality in Smart Grid - Power Quality Issues - Power Quality Conditioners for Smart Grid - Web Based Power Quality Monitoring - Power Quality Audit.					
UNIT - V	HIGH PERFORMANCE COMPUTING					9
	Basics of Communication Systems used in Smart Grid - LAN, HAN, WAN, Broadband Over Power Line (BPL), IP based Protocols - Basics of Web Service and Cloud Computing - Cyber Security for Smart Grid.					
	Total (L)					45 Periods
	* INTERNAL EVALUATION ONLY					
	OPEN-ENDED PROBLEMS / QUESTIONS					
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.					

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Explain smart grid concepts, features, and Indian smart grid initiatives.	L2 - Understand
CO2	Describe smart grid technologies and renewable energy integration methods.	L2 - Understand
CO3	Understand AMI systems and demand response programs.	L2 - Understand
CO4	Apply mitigation methods to solve power quality issues in smart grids.	L3 - Apply
CO5	Understand communication systems, cloud computing, and cyber security used in smart grid applications.	L2 - Understand
REFERENCE BOOKS:		
1.	Stuart Borlase "Smart Grid: Infrastructure, Technology and Solutions", CRC Press 2012.	
2.	James Momoh, "Smart Grid: Fundamentals of Design and Analysis", IEEE press, A John Wiley & Sons, Inc., Publication, 2012	
3.	Janaka Ekanayake, Nick Jenkins, Jianzhong Wu, Akihiko Yokoyama, "Smart Grid: Technology and Applications", Wiley, 2012.	
4.	Mini S. Thomas, John D McDonald, "Power System SCADA and Smart Grids", CRC Press, 2015	
5.	Kenneth C.Budka, Jayant G. Deshpande, Marina Thottan, "Communication Networks for Smart Grids", Springer, 2014	



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ME23PE504	SOLAR PV AND ENERGY STORAGE SYSTEMS		CP	L	T	P	C
			3	3	0	0	3
Type of Course	Open Elective						
Offering Dept.	M.E. – POWER ELECTRONICS AND DRIVES						
Programme & Branch	Common to M.E (ISE, AE and SE)					Version: 1.0	
Course Objectives:							
1.	To understand sunlight characteristics, semiconductor basics, and the working of solar cells.						
2.	To learn the design and components of stand-alone and hybrid solar PV systems.						
3.	To understand grid-connected PV systems, safety, performance, economics, and smart grid integration.						
4.	To understand the need for energy storage and different storage technologies for solar power.						
5.	To learn various practical applications of solar energy in industry, transport, and communication.						
UNIT-I	INTRODUCTION					9	
	Characteristics of Sunlight – Semiconductors and P-N Junctions – Behavior of Solar Cells – Cell Properties – PV cell Interconnection						
UNIT-II	STAND ALONE PV SYSTEM					9	
	Solar Modules – Storage Systems – Power Conditioning and Regulation – MPPT - Protection – Stand alone PV Systems Design – Hybrid PV Systems.						
UNIT-III	GRID CONNECTED PV SYSTEMS					9	
	PV Systems in Buildings – Design Issues for Central Power Stations – safety – Economic aspect – Efficiency and Performance - Smart Grid Integration.						
UNIT – IV	ENERGY STORAGE SYSTEMS					9	
	Impact of Intermittent Generation – Battery Energy Storage – Solar Thermal Energy Storage – Pumped Hydroelectric Energy Storage - Supercapacitors and Flywheel Energy Storage – Principles and Applications						
UNIT – V	APPLICATIONS					9	
	Solar Desalination - Water Pumping – Battery Chargers – Solar Car – Direct-drive Applications – Space – Telecommunications.						
						Total(L)	45 Periods
	* INTERNAL EVALUATION ONLY						
	OPEN-ENDED PROBLEMS / QUESTIONS						
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.						
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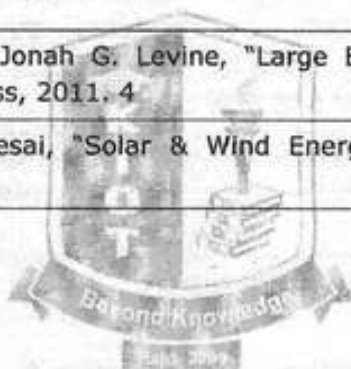
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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Explain the working principles of solar cells and PV systems.	L2 - Understand
CO2	Design basic stand-alone and hybrid solar PV systems.	L3 - Apply
CO3	Understand grid-connected PV systems and their integration with smart grids.	L2 - Understand
CO4	Identify suitable energy storage systems for intermittent solar generation.	L2 - Understand
CO5	Apply solar energy concepts to real-world applications.	L3 - Apply
	REFERENCE BOOKS:	
1.	Solanki C.S., "Solar Photovoltaics: Fundamentals, Technologies And Applications", PHI Learning Pvt. Ltd., 2015.	
2.	Stuart R.Wenham, Martin A.Green, Muriel E. Watt and Richard Corkish, "Applied Photovoltaics", 2007,Earthscan, UK. Eduardo Lorenzo G. Araujo, "Solar electricity engineering of photovoltaic systems", Progenisa,1994.	
3.	Frank S. Barnes & Jonah G. Levine, "Large Energy storage Systems Handbook", CRC Press, 2011. 4	
4.	McNeils, Frenkel, Desai, "Solar & Wind Energy Technologies", Wiley Eastern, 1990 5	



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UX23AC701	INDUSTRIAL SAFETY ENGINEERING	CP	L	T	P	C
		2	2	0	0	0
Type of Course	Audit Course					
Offering Dept.	Mechanical Engineering					
Programme & Branch	Common to M.E. (AE, ISE, PED and SE)					Version: 1.0
Course Objectives:						
1.	To provide foundational knowledge on the principles, history and global evolution of industrial safety.					
2.	To give understanding of air and water pollution and their control.					
3.	To understand the causes and effects of fire.					
4.	To understand the causes of accidents due to electrical hazards.					
5.	To enable students to have knowledge on improving road safety in relation with the driver.					
UNIT-I	INTRODUCTION TO INDUSTRIAL SAFETY					6
	Fundamentals of Safety and it's Management- Safety Planning and Risk Management - Accident Investigation and Prevention- Safety Performance Monitoring- Safety Education - Training and Employee Participation- Data Analysis for safety- Industrial Safety the Factories Act - Environmental Legislations and Waste Management Rules- International Standards.					
UNIT-II	ENVIRONMENTAL SAFETY					6
	Air Pollution- Air Quality Standards - Prevention and Control Measures - Water Pollution - Types of Water Pollutants (Organic, Inorganic, Microbial) - Treatment and Control Methods - Hazardous Waste Management - Treatment and Disposal Methods- Recycling and Recovery Techniques - Environmental Measurement and Control - Pollution Control in Process Industries.					
UNIT-III	FIRE SAFETY AND CHEMICAL SAFETY					6
	Fire Safety - Physics and Chemistry of Fire- Fire Prevention and Protection- Industrial Fire Protection Systems - Building Fire Safety - Explosion Protecting Systems. Chemical Safety - Hazard Identification (GHS / MSDS) -Acute and Chronic effects - Chemical Storage and Segregation -Handling, Transfer and PPE Requirements-Toxic Exposure and First Aid Measures.					

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UNIT – IV	ELECTRICAL SAFETY	6
	Concepts and Statutory Requirements - Electrical Hazards - Types of Electrical Hazards - Electric Shock and Burns - Protection Systems - Fuses and Circuit Breakers - Earth Fault Protection-Overload and Short-Circuit Protection-Selection and Installation-Operation and Maintenance - Preventive and Breakdown Maintenance - Lockout/Tagout Procedures - Hazardous zones.	
UNIT-V	TRANSPORT SAFETY	6
	Transportation of Hazardous Goods - Types of Hazardous Goods - Documentation, Regulatory Compliance- Road Transport - Driver Safety - Road Safety - Speed Control and Safe Distance - Accident Prevention Methods - Shop Floor and Repair Shop Safety.	
Total (L)		30 Periods
OPEN-ENDED PROBLEMS / QUESTIONS		
Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.		
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Explain the safety education and training.	L2-Understand
CO2	Illustrate and familiarize the basic concepts scope of environmental safety.	L2-Understand
CO3	Apply explosion protection techniques and their significances to suit the Industrial requirement.	L3 – Apply
CO4	Choose various protection systems from different electrical operations.	L3 – Apply
CO5	Apply road safety principles, including accident prevention, Motor Vehicles Act, insurance, and safety surveys to improve transport safety.	L3 – Apply
REFERENCE BOOKS:		
1.	Hughes. P & Ferrett. E, "Introduction to health and safety at work", Routledge.	
2.	The Factories Act, 1948, Madras Book Agency.	
3.	T. Miller, "Environmental Science: Working with the Earth", 11th Edition, Wadsworth Publishing Co., Belmont, CA, 2006.	

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4.	"Fire Prevention and firefighting", Loss prevention Association, India.
5.	Indian Electricity Act and Rules, Government of India.
6.	K.W.Ogden, "Safer Roads – A guide to Road Safety Engineering".




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UX23AC702		SUSTAINABILITY ENGINEERING				CP	L	T	P	C
						2	2	0	0	0
Type of Course	Audit Course									
Offering Dept.	Mechanical Engineering									
Programme & Branch	Common to M.E (AE, ISE, PED and SE)								Version: 1.0	
Course Objectives:										
1.	To develop an understanding of sustainability principles and their application in engineering design and practice.									
2.	To understand the impacts of greenhouse gas emissions on global warming and climate change.									
3.	To understand and evaluate strategies for enhancing water conservation and minimizing water wastage.									
4.	To understand the relationship between air quality and human health and the environmental impacts of air pollution.									
5.	To examine governance challenges associated with effective supplier and partner engagement.									
UNIT-I	INTRODUCTION TO SUSTAINABILITY ENGINEERING								6	
	Definition and principles of sustainability-History and evolution of sustainability-Growth into the mainstream - Sustainability in engineering practice - Sustainable Development Goal: (SDGs)- Challenges and opportunities in sustainability engineering. * Experiential Learning: Zero Waste Campus Challenge. Idea: Develop a full segregation and composting system. Introduce "Green Points" for departments reducing single-use plastics.									
UNIT-II	GREENHOUSE GAS EMISSION AND CLIMATE CHANGE								6	
	Introduction of Green House Gas (GHG) - Major GHGs- GHG Effects-Realization of climate action: BIRTH OF United Nations Framework Convention on Climate Change (UNFCCC) - Carbon Border Adjustment Mechanism (CBAM) - ISO 14064 series- Scope 1,2&3- GHG Protocol - GHG accounting and reporting principles - Emission calculation - Global Warming Potential (GWP)-Reporting procedure-Mandatory requirement. * Experiential Learning: CO2 Capture through Green Walls and Algae Bio-Reactors. Idea: Design vertical gardens and small algae ponds near high-traffic areas (canteen, generator room) to absorb CO2 and improve air quality.									

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UNIT-III	ENVIRONMENTAL STEWARDSHIP: WATER, AIR AND LAND	6
	Water Stewardship: Managing freshwater - Groundwater- Oceans through rainwater harvesting - Preventing pollution. Air Stewardship: Reducing air pollution - Improving air quality. Land Stewardship: Responsible land use - Preventing erosion- Improving soil fertility - Managing land for multiple uses: Biodiversity, Community and Economy. * Experiential Learning: Campus Water Neutrality Project. Idea: Measure total water consumption vs. recharge. Implement rainwater harvesting, greywater reuse (from hostels/labs), and smart metering.	
UNIT - IV	CORPORATE SOCIAL RESPONSIBILITY	6
	Concept of Corporate Social Responsibility (CSR) - Principles of CSR: Accountability, Transparency, Ethical behavior - Respect for stakeholder interests - Respect for rule of law - Respect for international norms - Respect for human rights - Guidelines on CSR - Impact of CSR-CSR community partnership-ISO 26000-ESG reporting - key features. * Experiential Learning: ESG Student Ambassador Program. Idea: Form a student-led ESG club responsible for awareness, campaigns, and behavior-change initiatives like "bike-to-cycle".	
UNIT-V	GOVERNANCE, RISK MANAGEMENT AND ETHICS	6
	Conceptual framework of corporate governance - Board effectiveness - Corporate policies & disclosures - Shareholders' rights- Risk identification, mitigation and audit - Compliance management - Ethics & Business - Indian and contemporary laws relating to anti-bribery- Global report. *Experiential Learning: Green Procurement and Ethical Supply Chain Policy. Idea: Develop campus policy for sustainable purchases - Eco-stationery, energy-efficient equipment and recycled materials.	
	Total (L)	30 Periods
	* INTERNAL EVALUATION ONLY	
	OPEN-ENDED PROBLEMS / QUESTIONS	

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	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
C01	Explain the challenges and opportunities involved in sustainability engineering practices.	L2- Understand
C02	Apply GHG accounting and reporting boundaries and to prepare a basic GHG inventory.	L3 – Apply
C03	Discuss principles of land stewardship, including responsible land use, erosion control, and improvement of soil fertility.	L3 – Apply
C04	Analyze the impact of CSR initiatives on business performance, reputation, and sustainable development outcomes.	L3 – Apply
C05	Outline processes for risk identification, mitigation, internal control, audit, and compliance management within a corporate governance system.	L3 – Apply
	REFERENCE BOOKS:	
1.	Wahidul K. Biswas, Michele John, "Engineering for Sustainable Development: Theory and Practice", Wiley Publishers, 2022.	
2.	R. L. Rag and Lekshmi Dinachandran Remesh, "Introduction to Sustainable Engineering", 2 nd Edition, PHI Learning Pvt. Ltd., 2016.	
3.	Bhavik R. Bakshi, "Sustainable Engineering: Principles and Practice", Cambridge University Press, 2019.	
4.	A.S. Bradley, A. O. Adebayo, P.Maria, "Engineering applications in sustainable design and development", 1 st Edition, Cengage learning, 2016.	

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Faculty of CSE & IT
Knowledge Institute of Technology
KIOT Campus, Kakapalayam
Salem-637 504

M.E. / M.Tech. / M.C.A. Regulations 2023

UX23AC703	ENERGY AUDIT		CP	L	T	P	C
			2	2	0	0	0
Type of Course	Audit Course						
Offering Dept.	Electrical and Electronics Engineering						
Programme & Branch	Common to M.E (ISE, AE, PED and SE)					Version: 1.0	
Course Objectives:							
1.	To understand India's energy scenario, energy audits, and energy policies.						
2.	To learn audit types, methods, cost analysis, and audit tools.						
3.	To learn energy management programs, audit reports, and energy savings.						
4.	To understand thermal energy saving methods in industrial and building systems.						
5.	To learn efficient electrical energy use and conservation techniques.						
UNIT-I	INTRODUCTION					6	
	Review of Energy Scenario In India - General Philosophy and Need of Energy Audit and Management - Basic Elements and Measurements - Mass and Energy Balances - Scope of Energy Auditing Industries - Evaluation of Energy Conserving Opportunities - Energy Performance Contracts, Need for Energy Policy for Industries, National & State level Energy policies.						
UNIT-II	ENERGY AUDIT METHODOLOGIES					6	
	Types of Energy Audit - Energy Management (audit) Approach - Understanding Energy Costs - Benchmarking - Matching Energy use to Requirement - Optimizing the input Energy Requirements - Duties and Responsibilities of Energy Auditors - Energy Audit Instruments.						
UNIT-III	ENERGY MANAGEMENT					6	
	Design of Energy Management Programmes - Development of Energy Management Systems - Importance - Duties of Energy Manager - Energy Audit Reports - Case Study and Potential Energy Savings. * Experiential Learning: Preparation and presentation of Energy Audit Reports.						
UNIT - IV	THERMAL ENERGY MANAGEMENT					6	
	Energy Conservation in Boilers - Steam Turbines and Industrial Heating systems - Cogeneration and Waste heat recovery - Thermal Insulation - Heat Exchangers and Heat Pumps - Building Energy Management.						
UNIT - V	ELECTRICAL ENERGY MANAGEMENT					6	
	Methods to Minimize Supply-demand Gap - Renovation and Modernization of Power Plants - Reactive Power Management - Conservation in Motors, Pumps and Fan Systems - Energy Efficient Motors.						
					Total(L)		30 Periods

w.e.f. 22/12/2025

Passed in BoS Meeting held on 06/12/2025

Approved in Academic Council Meeting held on 15/12/2025

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M.E./M.Tech./M.C.A Regulations-2023

	* INTERNAL EVALUATION ONLY	
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Explain India's energy scenario, energy audits, and energy policies.	L2 - Understand
CO2	Apply energy audit methods and benchmarking techniques.	L3 - Apply
CO3	Develop basic energy management programs and audit reports.	L3 - Apply
CO4	Identify thermal energy-saving opportunities in industrial systems.	L2 - Understand
CO5	Apply electrical energy conservation techniques in power systems and equipment.	L3 - Apply
	REFERENCE BOOKS:	
1.	De, B. K., "Energy Management, Audit & Conservation", 2nd Edition, Vrinda Publication, 2010	
2.	Murphy, W. R., "Energy Management", 1st Edition, Elsevier India Private Limited, 2007	
3.	Turner, W. C., Doty, S., Turner, W. C., "Energy Management Handbook", 7th Edition, Fairmont Press, 2009	
4.	Patrick, F., "Energy Conservation Guidebook", 1st Edition, Prentice Hall, 1993	

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Salem-637 504

UX23AC704	DESIGN THINKING	CP	L	T	P	C
		2	2	0	0	0
Type of Course	Audit Course					
Offering Dept.	Computer Science and Engineering					
Programme & Branch	Common to M.E. (ISE, AE, PED, SE)				Version: 1.0	
Course Objectives:						
1.	To learn design thinking concepts and principles.					
2.	To use design thinking methods in every stage of the problem.					
3.	To learn the different phases of design thinking.					
4.	To develop a prototype and perform testing.					
5.	To understand the character and quality of an entrepreneur.					
UNIT-I	INTRODUCTION				6	
	Need for Design - Four Questions - Ten Tools - Principles of Design Thinking - The process of Design Thinking.					
UNIT-II	UNDERSTAND, OBSERVE AND DEFINE THE PROBLEM				6	
	Understanding of the problem - Problem analysis - Reformulation of the problem - Observation Phase - Empathetic design - Characterization of the target group - Description of customer needs.					
UNIT-III	IDEATION AND PROTOTYPING				6	
	Ideate Phase - Evaluation of ideas - Prototype Phase - Lean Startup Method for Prototype Development - Visualization and presentation techniques.					
UNIT-IV	TESTING AND IMPLEMENTATION				6	
	Test Phase - Tips for interviews - Tips for surveys - Kano Model - Desirability Testing - Conducting workshops - Requirements for the space.					
UNIT-V	ENTREPRENEURSHIP				6	
	Entrepreneurship - Character - Quality of Entrepreneur - Opportunity - Entrepreneurial design thinking - The New Social Contract - Design Activism - Designing tomorrow.					
				Total (L)		30 Periods

w.e.f. 22/12/2025

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Describe key concepts of design thinking.	L2 – Understand
CO2	Discuss the phases of design thinking process.	L2 – Understand
CO3	Illustrate design thinking in all stages of problem solving.	L3 – Apply
CO4	Apply testing methodologies to validate the prototype.	L3 – Apply
CO5	Determine the role of an entrepreneur.	L3 – Apply
REFERENCE BOOKS:		
1.	Christian Mueller-Rotenberg, "Handbook of Design Thinking - Tips & Tools for how to design thinking", 2018.	
2.	Jeanne Liedtka and TimOgilvie, "Designing for Growth: A Design Thinking Tool Kit for Managers", Columbia University Press, 2011	
3.	Tim Brown, "Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation", HarperCollins e-books, 2009.	



w.e.f. 22/12/2025
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UX23AC705	BUSINESS ANALYTICS	CP	L	T	P	C
		2	2	0	0	0
Type of Course	Audit Course					
Offering Dept.	Computer Science and Engineering					
Programme & Branch	Common to M.E. (ISE, AE, PED, SE)				Version: 1.0	
Course Objectives:						
1.	To understand the analytics life cycle.					
2.	To comprehend the process of acquiring business intelligence.					
3.	To understand various types of analytics for business forecasting.					
4.	To model the supply chain management for analytics.					
5.	To apply analytics for different functions of a business.					
UNIT-I	INTRODUCTION TO BUSINESS ANALYTICS					6
	Analytics and Data Science - Analytics Life Cycle - Types of Analytics - Business Problem Definition - Data Collection - Data Preparation - Hypothesis Generation - Modeling - Validation and Evaluation - Interpretation - Deployment and Iteration.					
UNIT-II	BUSINESS INTELLIGENCE					6
	Data Warehouses and Data Mart - Knowledge Management -Types of Decisions - Decision Making Process - Decision Support Systems - Business Intelligence - OLAP - Analytic functions.					
UNIT-III	BUSINESS FORECASTING					6
	Introduction to Business Forecasting and Predictive analytics - Logic and Data Driven Models - Data Mining and Predictive Analysis Modelling - Machine Learning for Predictive analytics.					
UNIT-IV	HR & SUPPLY CHAIN ANALYTICS					6
	Human Resources - Planning and Recruitment - Training and Development - Supply chain network - Planning Demand, Inventory and Supply - Logistics - Analytics applications in HR & Supply Chain - Applying HR Analytics to make a prediction of the demand for hourly employees for a year.					
UNIT-V	MARKETING & SALES ANALYTICS					6
	Marketing Strategy - Marketing Mix - Customer Behaviour - selling Process - Sales Planning - Analytics applications in Marketing and Sales - predictive analytics for customers' behaviour in marketing and sales.					
					Total (L)	30 Periods

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Explain the real world business problems and model with analytical solutions.	L2 – Understand
CO2	Identify the business processes for extracting Business Intelligence	L2 – Understand
CO3	Apply predictive analytics for business fore-casting	L3 – Apply
CO4	Apply analytics for supply chain and logistics management	L3 – Apply
CO5	Use analytics for marketing and sales.	L3 – Apply
REFERENCE BOOKS:		
1.	R. Evans James, "Business Analytics", 2 nd Edition, Pearson, 2017.	
2.	R.N. Prasad, Seema Acharya, "Fundamentals of Business Analytics", 2 nd Edition, Wiley, 2016.	
3.	Philip Kotler and Kevin Keller, "Marketing Management", 15 th edition, PHI, 2016.	
4.	VSP RAO, "Human Resource Management", 3 rd Edition, Excel Books, 2010.	
5.	Mahadevan B, "Operations Management -Theory and Practice", 3 rd Edition, Pearson Education, 2018.	



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